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Parenteral Drug Association

Technical Report No. 46 (Revised 2024)

Last Mile: Guidance for Good Distribution Practices for Pharmaceutical Products to End Users

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Last Mile: Guidance for Good Distribution Practices for Pharmaceutical Products to End Users

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1.0 Introduction

This guidance document complements the information provided in *PDA Technical Report No. 39 (Revised 2021): Guidance for Temperature-Controlled Medicinal Products — Maintaining the Quality of Temperature-Sensitive Medicinal Products through the Transportation Environment* by going beyond manufacturer-focused issues and addressing issues specifically encountered within the "last mile" of the distribution of medicinal products and devices *(1)*. This revision updates the original 2009 technical report to account for expanded regulations, the development and implementation of newer, more advanced technologies and supply chains that have become more extensive and complex *(2)*.

1.1 Purpose and Scope

"Last mile" distribution is an often-used logistical term that refers to the transport of product from the manufacturing facility to the end user. For the purposes of this technical report, last mile is defined as the final transfer of a drug from the manufacturer or wholesaler to the end user which could be either the patient or the entity (e.g., pharmacy, hospital) that provides the drug to the patient. Transfers earlier in the distribution process involve organizations with greater knowledge of the product and the resources to ensure carefully controlled conditions of storage and transport compared to those available to the end user. As such, PDA *Technical Report No. 46 (Revised 2024): Last Mile: Guidance for Good Distribution Practices for Pharmaceutical Products to End User* provides recommendations and tools based on current best practices to ensure that product quality is maintained until the final handoff.

2.0 Glossary and Abbreviations

The Glossary includes a list of common terms specific to this technical report. Definitions of terms are listed alphabetically and may come from sources such as the U.S. Pharmacopeia, American Society for Testing Materials, International Conference for Harmonisation, International Organization for Standardization, or other PDA technical reports. Definitions quoted from other sources are cited at the end of the definition.

ALCOA+

ALCOA+ is an acronym standing for Attributable, Legible, Contemporaneous, Original, Accurate ("ALCOA") and Complete, Consistent, Enduring and Available ("+"). It is a set of characteristics that all data must meet to ensure data integrity. Adherence to ALCOA+ principles is the basis for Good Documentation Practice.

Corrective Action, Preventive Action (CAPA)

Action to eliminate the cause of a detected nonconformity or other undesirable situation. Corrective action is taken to prevent recurrence, whereas preventive action is taken to prevent occurrence.

Customer

In distribution, the customer is the trading partner or reseller; in direct-to-consumer situations, the customer is the end user who is either the patient taking the product or the person administering the product to the patient such as a physician.

Data Integrity

Data integrity is the degree to which data are complete, consistent, accurate, trustworthy, reliable and that these characteristics of the data are maintained throughout the data life cycle *(3)*.

Downstream

Refers to the demand side of the supply chain that consists of one or more companies or individuals who participate in the flow of pharmaceuticals moving from the manufacturer to the end user.

Good Distribution Practices (GDP)

Defined as the part of quality assurance that ensures the quality of the pharmaceutical

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product is maintained by means of adequate control of the numerous activities that occur during the distribution process.

Good Manufacturing Practices (GMP)

Best practices in manufacturing of pharmaceuticals or biopharmaceuticals, from a regulatory standpoint, are regarded as the minimum current good manufacturing practice for methods to be used in, and the facilities or controls to be used for, the manufacture, processing, packaging, or holding of a drug to ensure it meets requirements of safety, identity, and strength and meets the quality and purity characteristics that it purports or is represented to possess.

Last Mile

Last mile is a multidimensional, cross-industry term that defines a point in the supply chain where the product or service directly faces the customer, end user, or patient.

Mean Kinetic Temperature (MKT)

The single calculated temperature at which the total amount of degradation over a particular period is equal to the sum of the individual degradations that would occur at various temperatures. Thus, MKT may be considered as an isothermal storage temperature that simulates the nonisothermal effects of storage temperature variation. It is not a simple arithmetic mean.

Metadata

Metadata are data that describe the attributes of other data and provide context and meaning. Typically, these are data that describe the structure, data elements, inter-relationships, and other characteristics of data, e.g., audit trails. Metadata also permit data to be attributable to an individual (or if automatically generated, to the original data source) *(3)*.

Original Record

Data as the file or format in which it was originally generated, preserving the integrity (accuracy, completeness, content and meaning)

of the record, e.g., original paper record of manual observation, or electronic raw data file from a computerized system *(3)*.

Pedigree, e-Pedigree

A pedigree is a record, containing information regarding each transaction, resulting in a change of ownership of a pharmaceutical, from sale by a manufacturer, through acquisition and sale by a wholesaler, until final sale to a pharmacy or person furnishing, administering, or dispensing the pharmaceutical.

Phase Change Material (PCM)

A physical material that stores and releases thermal energy when freezing or melting. A PCM releases energy when freezing [latent heat energy] and absorbs energy when melting.

Radio Frequency Identification (RFID)

Is an automatic technique for identifying objects using radio frequency transmissions. An RFID system generally consists of a tag, reader, antenna, and software. An RFID tag is simply another type of data carrier. Essentially, tags comprise a semiconductor chip with memory, processing capability and a transmitter connected to an antenna (aerial).

Reverse Logistics

The process of planning, implementing, and controlling the efficient, cost-effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or disposal.

Shipping Lane

An established route of transportation between a specific point of origin to a specific destination. More defined routes will consistently use the same combination and types of transportation vehicles and transfer procedures, but this may not be observed through all shipping lanes throughout the last mile.

Supply Chain

The chain of events or process which includes all storage and movement of raw materials, work-in-process inventory, and finished goods from point-of-origin to point-of-consumption.

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True Copy

An exact copy of an original recording of data that has been verified and certified to confirm it is an exact and complete copy that preserves the entire content and meaning of the original record, including, in the case of electronic data, all essential metadata and the original record format as appropriate (can be referred to also as certified copy) *(4)*.

Upstream

Refers to the supply side of the supply chain. Upstream partners are the suppliers who provide goods and services to the organization that are to satisfy demands, which originate at the point of demand or use, as well as other flows, such as return product movements, payments for purchases, etc. This is the opposite of downstream.

2.1 Abbreviations

EPR Extended Product Responsibility

FEFO First Expired, First Out

GDP Good Distribution Practices

GDocP Good Documentation Practices

GDPR General Data Protection Regulation

IoT Internet of Things

MAH Marketing Authorization Holder

MKT Mean Kinetic Temperature

NGO Non-governmental Organizations

RFID Radio Frequency Identification

3PL Third-Party Logistic Provider

3.0 Pharmaceutical Supply Chains

The pharmaceutical supply chain has a high level of complexity as it varies regionally and in the number of stakeholders including but not limited to manufacturers, distribution centers, wholesalers, points of dispensing, and points of administration. In addition, many countries have public and private pharmaceutical supply chains managed by governmental organizations and wholesale distributors respectively. Before a patient receives a pharmaceutical product, distribution routes should be appropriately managed in the supply chain to maintain product safety, integrity, quality, and efficacy. Therefore, the roles and responsibilities of supply chain partners and the design of the supply chain should be clearly determined to manage the environmental conditions, regulations, and product requirements for delivery to patients.

3.1 Distribution in Supply Chains

Distribution of medicinal products in the supply chain can be divided into internal and external distribution. Internal distribution can be considered as the transport of a commercial product within a pharmaceutical manufacturer's supply chain; that is, all transports from the manufacturing facility to the packaging facility to the warehouse to a distribution center *(5)*. During internal distribution, the marketing authorization holder (MAH) is the owner of the goods at all times. An internal distribution network with direct distribution to the customer (also called direct sales) is subject to review by inspectorates as part of the manufacturer's good manufacturing practices (GMP) and good distribution practices (GDP) authorization. Internal distribution is typically called the "first mile" and direct distribution from the manufacturer to a point of administration (e.g., hospital) is referred to as the "last mile." In some unique circumstances, manufacturers may ship directly to the patient such items as test kits and orphan drugs, for example, in compliance with local laws and regulations. When the manufacturer is shipping directly to the patient, the last-mile best practices outlined in this report should be considered and followed.

External distribution, on the other hand, is product that is no longer owned by the manufacturer. It covers the transport of commercial products through various steps in the customer's supply chain, that is, transport from a pharmaceutical manufacturer's distribution center to commercial customers

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(including wholesalers, retailers, and buying groups like non-governmental organizations (NGO)), clinical facilities, or directly to the patient. External distribution is normally not part of a product filing, and it can be called "middle mile" as well as last mile to the end user the patient. Here the last mile is the final transportation path to product administration. This can be the product pick-up from a point of dispensing, such as a pharmacy to a patient's home or a shipment of goods from a (wholesaler) warehouse to a point of administration like a hospital or vaccine center.

Moving through the supply chain, product volumes decrease from product-pallet shipments in the first mile to single packs in the last mile. Meanwhile, the mix of products typically varies in a shipment from one or a few product(s) in the first mile to multiple or more than hundred products in the middle mile to one or a few products in the last mile. In addition, the distance and duration of the shipments typically decrease from thousands of miles (from continent to continent) and multiple transportation days by air and ocean in the first mile to a few kilometers and hours or minutes in the last mile. Exceptions do exist, however; for example, the last mile can take days in rural and remote areas where transportation by foot or via rivers by boat is the only option.

Within the external distribution network, there may be two supply chains for pharmaceuticals—public and private. Public supply chains are owned by the government and focus on essential medicines for the population, like vaccines and products for the prevention or treatment of malaria and human immunodeficiency viruses. Very often NGOs donate products to public supply chains to support the population in low-income and/or developing countries. The public supply chains typically comprise central, regional, and district government distribution centers, public pharmacies, health centers, and public hospitals. Private supply chains typically comprise wholesalers, pharmacies, clinics, and healthcare professionals authorized to dispense medicinal products to the public. In this case, wholesalers also deliver goods to hospitals and health centers via their network of distribution centers.

The external distribution network is overseen by inspectorates as part of the wholesaler's GDP authorization. **Figure 3.1-1** summarizes, at a high level, pharmaceutical supply chains, the relationship between internal and external distribution, and the differences between first-, middle-, and last-mile distribution. In some cases, or in some countries, a pharmaceutical manufacturer can also own a local wholesaler distribution license, thereby owning product and taking full responsibility for the distribution network in the middle and the last miles.

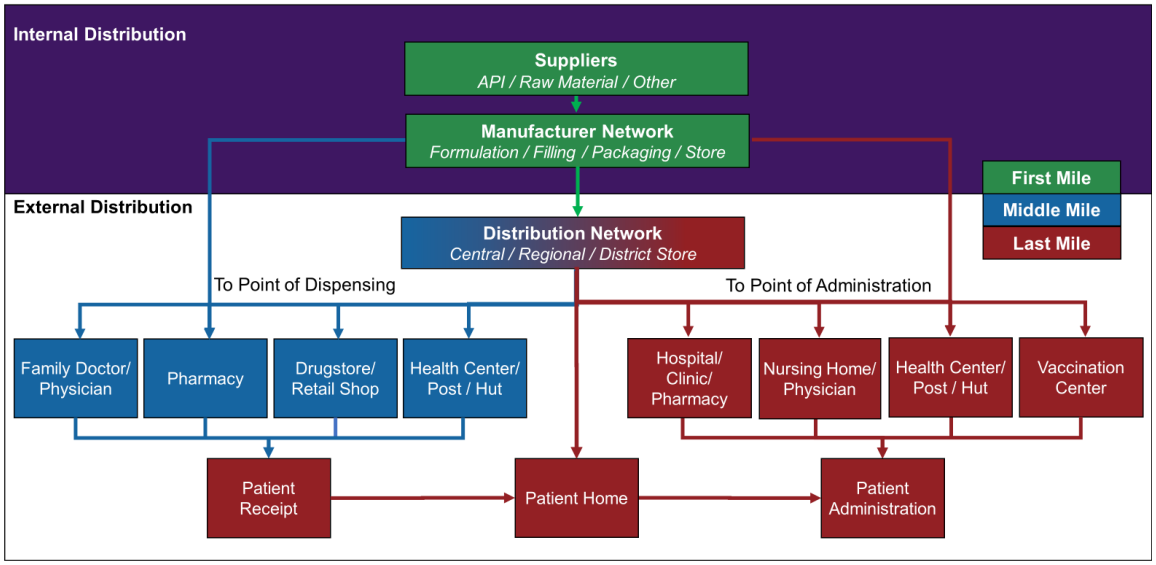

Figure 3.1-1 Schematic Representation of the Pharmaceutical Supply Chain

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3.2 Roles and Responsibilities in Supply Chains

There are a number of discernable roles and responsibilities in the pharmaceutical supply chain:

Manufacturer

Produces goods in the internal distribution network and executes stability testing to set expiry, storage, and transportation conditions.

Wholesaler

Procures, holds, supplies, imports, or exports medicinal products in an external private distribution network, apart from supplying medicinal products to the public.

Public Medical Store

Stores and ships goods in a public distribution network along the health pyramid.

Non-Governmental Organization

Professional organization that donates goods to public supply chains.

Service Delivery Point (SDP)

Receives goods that are supplied to patients.

Point of Administration

Receives goods that are administrated to patients, e.g., hospital, clinic, vaccine center.

Point of Dispensing

Location where goods are dispensed to users, e.g., patient.

Marketing Authorization Holder

The person or company who is licensed to distribute, sell, and commercialize a medical product.

Third-Party Logistic Provider (3PL) or **Logistic Service Provider**

Entity that, on behalf of a manufacturer or wholesaler, executes logistics activities like warehousing and transportation in the supply chain.

Customs

The official department that administers and collects the duties levied by the government on imported goods.

Insurance Company

A term insurance company, as its primary and predominant business activity during the taxable year, issues insurance or annuity contracts or reinsures the risks underwritten by other insurance companies.

Quality Assurance

Ensures that a quality system is in place to guarantee product is being distributed in compliance with GDP and that the quality, safety, and efficacy of the pharmaceutical product is maintained.

Pharmaceutical Products Destruction Commission

Multisectoral committee oversees the destruction of expired or deteriorated pharmaceutical products in accordance with hygiene laws and regulations.

Inspectorate

Inspects manufacturers, wholesalers, pharmacies, or processors of regulated products to verify compliance with relevant regulations and provides permits to the MAH to import pharmaceutical products into a specified market.

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All participants in the supply chain are responsible for maintaining product quality and the right storage and transportation conditions for the delivery of effective and safe products to patients. Communication about product quality requirements for all players is essential to achieve this aim. Defining the shared responsibility of the participants in a written Quality Agreement is a good practice.

Section 3.3 through **Section 3.9** provide examples of pharmaceutical supply chains in various regions around the world.

3.3 Supply Chains in Europe

Most pharmaceuticals in the European Union (EU) are distributed via private supply chains, while the public supply chains are used primarily to distribute vaccines to vaccine centers for vaccination programs. In addition, NGOs buy pharmaceutical products from manufacturers for donation programs that they then provide to public supply chains on other continents like Africa and Asia. The *Guidelines of 5 November 2013 on Good Distribution Practice of Medicinal Products for Human Use* are the most important GDP guidelines in the EU: they are in force in all EU member states and in several other countries as well. The GDP guidelines of the non-EU member states such as Russia and the United Kingdom share many similarities with the EU GDP guidelines.

The EU GDP guidelines require a wholesaler to be licensed to distribute an approved product into the region in which the wholesaler will sell that product. While there are many actors in the supply chain, in the EU, almost 60% of all medicinal products sold are distributed to pharmacies through pharmaceutical wholesale distributors, while 33% are distributed by the manufacturer directly to hospitals and 8% directly to pharmacies *(6)*.

Pharmaceutical full-line wholesaling activity consists of the purchase, warehousing, storage, order preparation, and delivery of medicines. Pharmaceutical full-line wholesalers carry and distribute a complete range and depth of products within the framework set by the authorities and the market to meet the needs of those with whom they have normal business relations. They also have the capability to deliver all medicines in their geographical area of activity typically on the same day or within 24 hours on working days. Pharmaceutical full-line wholesalers provide working capital, extend financing services, and fund stock and receivables of pharmacies and healthcare professionals. These factors distinguish them from other distributors/wholesalers in the supply chain.

In most EU member states, pharmaceutical full-line wholesalers also respond to Public Service Obligations or carry a Public Service Function as charged by Article 81 of Directive 2004/27/EC. Article 81 introduced an obligation for member states to implement the following measures:

> The holder of a marketing authorization for a medicinal product and the distributors of the said medicinal product actually placed on the market in a Member State shall, within the limits of their responsibilities, ensure appropriate and continued supplies of that medicinal product to pharmacies and persons authorized to supply medicinal products so that the needs of patients in the Member State in question are covered *(7)*.

Apart from this provision, many EU member states (e.g., Belgium, France, Germany, Greece, Italy, Slovenia, Spain, Portugal) have appended a separate Public Service Obligation on wholesale distributors to guarantee that certain products be made available across the region. This ensures that through a permanently available and adequate range of medicinal products, referred to as buffer stocks, the requirements of any specific geographical area will be met, and requested medicinal products can be delivered in a timely manner. In addition, one of the minimum requirements for wholesalers of medicinal products to obtain their distribution authorization (as set by Article 84 of Directive 2001/83/EC), is compliance with the principles and guidelines of GDP *(8)*.

The EU GDP guidelines also address the issue of handling and transporting temperature-sensitive medicinal products. The general requirement of suitability does not specifically target temperature-

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sensitive products but requires that "[w]holesale distributors must have suitable and adequate premises, installations and equipment, so as to ensure proper storage and distribution of medicinal products. In particular, the premises should be clean, dry and maintained within acceptable temperature limits…." (Sec. 3.1.) and "[t]he premises should be designed or adapted to ensure that the required storage conditions are maintained…." (Sec. 3.2). Furthermore, "[s]uitable equipment and procedures should be in place to check the environment where medicinal products are stored. Environmental factors to be considered include temperature, light, humidity and cleanliness of the premises…." (Sec. 3.2.1).

In general, a guiding principle underpinning operations and actions undertaken by wholesale distributors is that "[t]he wholesale distribution of medicinal products is performed according to the information on the outer packaging" (Sec. 5.1) *(9)*.

The chapter of the EU GDP on transportation specifically targets temperature-sensitive products insofar as "[t]he required storage conditions for medicinal products should be maintained during transportation within the defined limits as described by the manufacturers or on the outer packaging" (Sec. 9.2). In the event of a temperature excursion, the incident should be reported to the distributor and recipient of the affected product. In that regard, it must be emphasized that the information on temperature control originates from either specific information from the manufacturer or the information on the outer packaging of the product. In any case, a risk-based approach should be used when planning transportation. As such, and because there is no further precision or requirement of stability data, any deviation of the temperature indications must be reported, in compliance with EU GDP.

Although the general regulatory framework laid down by Directive 2001/83/EC and the EU GDP guidelines originate at the EU level, member states are responsible for implementing these rules: "checks on the persons authorized to engage in activity as a wholesaler in medicinal products, and the inspection of their premises, shall be carried out under the responsibility of the Member State which granted the authorization for premises located on its territory" (Art. 1) *(8)*. These provisions are binding in the EU and are implemented in all member states. Each member state will entrust one or more competent authorities, to carry out inspections on the premises of wholesalers to check their compliance with the Directive and GDP.

Wholesale distribution authorizations and GDP certificates are available in a central database called EudraGMDP, the content of which is provided by the National Competent Authorities of the European Economic Area and is maintained and operated by the European Medicines Agency (EMA) *(10)*.

3.4 Supply Chains in Africa

Many African countries have adopted the World Health Organization (WHO) GDP guidance, applying Annex 5, Annex 7, and Annex 9, often in conjunction with local GDPs *(5, 11, 12)*. For example, Zambia, Kenya, Ethiopia, Nigeria, and Uganda have developed country-specific GDP guidelines and the Union Economique et Monetaire Ouest Africanine has established GDP guidelines for French-speaking African countries *(13-18)*. In Africa, many countries import about 90% of their pharmaceutical products, and customs and the inspectorate play a major role in the lead time needed to deliver products to the central warehouses, both in the private and public supply chains.

The African continent faces major challenges to delivering pharmaceuticals of the right quality and quantity to the patient, at the right place, and in a timely manner. Some of these challenges include the poor infrastructure of motorways, vehicle conditions, high temperatures and humidity, dry and rainy seasons, and power outages that impact temperature-controlled stores. The limited financial resources and lack of local manufacturers can create a dependency on donations from Faith-based Organizations (FBO) and NGOs for the import of medicines and, in some countries, poor supervision from health inspectorates and unstable political and governmental circumstances stand in the way.

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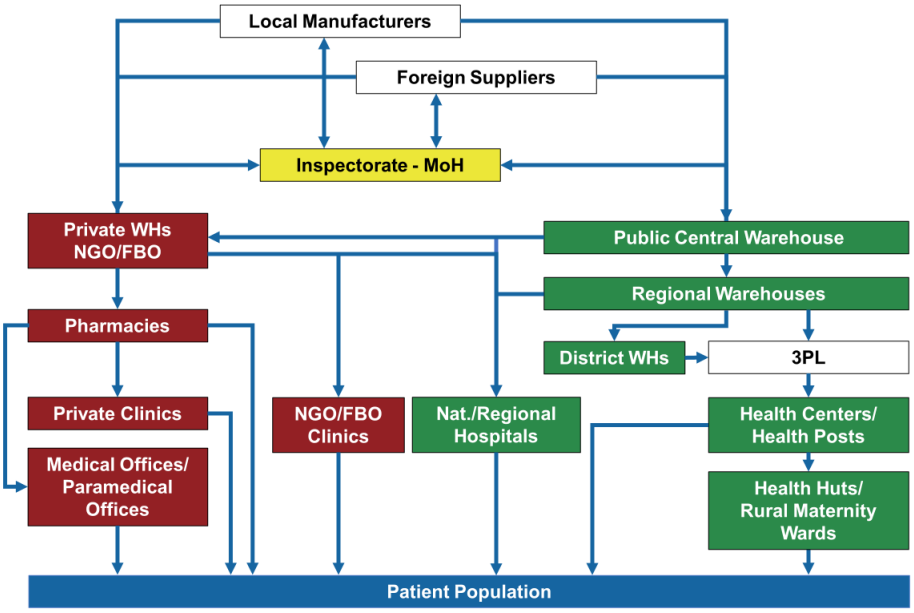
In Africa, the population has relatively less to spend on medication compared to other continents, and reimbursement for the cost of medications by insurance companies is limited. Therefore, the public pharmaceutical supply chains play a larger role in supplying medication for free to the population in Africa than they do on other continents. Public supply chains typically deliver medication to the population for HIV/AIDS, tuberculosis, malaria, nutrition, reproductive health, hepatitis, and noncommunicable diseases, while private supply chains provide other medications.

Public supply chains for health products in Africa are relatively long and generally follow the four- or five-level health system architecture (see **Figure 3.4-1**). Typically, a central warehouse distributes the goods to regional warehouses in the province. From the regional warehouse, goods are shipped to district warehouses/stores, and subsequently the goods are shipped via the district warehouses to the health facilities to reach the patient. In these supply chains, it may take 6 to 18 months to move goods from the central warehouse to the health facilities.

Given the complexity of the distribution networks, the procedures generally necessitate maintaining security stocks of one to two months at the SDP level, three months at the health district level, and four to six months at the regional and central levels. As a consequence, out-of-stock situations have a major impact on patient health, but the financial consequences of overstocking with safety stocks must also be considered.



Note: Red boxes show the private supply chain, green boxes display the public supply chain.

Figure 3.4-1 Example of Pharmaceutical Supply Chain in Senegal

In contrast, the public supply chain in Zambia was reorganized recently from a Tier-3 distribution network to a more last-mile model wherein the storage capacity at central and regional warehouses was expanded and all district warehouses were eliminated. A central information technology inventory management system for stock and purchase-order management and visibility was implemented for the entire supply chain, and a last-mile distribution system was established from the central warehouse and six regional warehouses to 3400 health facilities which included hospitals, clinics, centers, posts, and huts. In addition, regional warehouses are no longer present in each province; instead, their locations are based on population distribution, geographical environment, and distance to health facilities

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(allowing for a maximum eight-hour drive). As a result, the distance between distribution points is shorter than before, total inventory stock level is lower, and medicine accessibility has been improved.

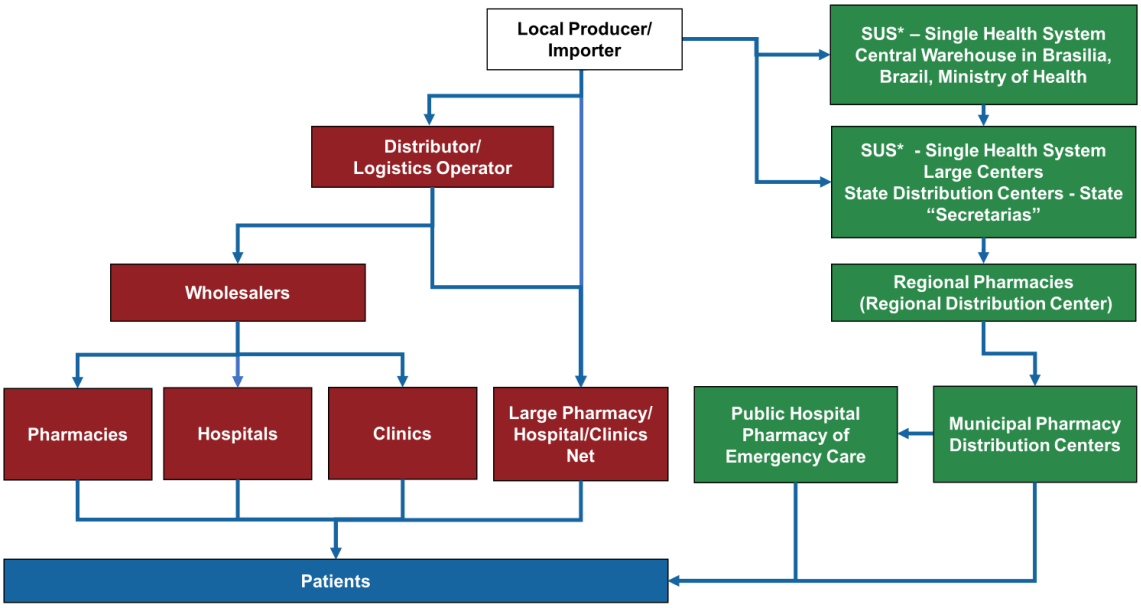
3.5 Supply Chains in Brazil

In Brazil, wholesalers, logistic service companies, distributors, pharmacies, clinics, and hospitals must be licensed to distribute, store, commercialize, and administrate an approved product, according to the *Portaria CVS 1/2020* as shown in **Figure 3.5-1** ***(19)*****.** The MAH may engage a third-party company for the distribution or logistics services, according to *Resolution of the Collegiate Board (RDC) n° 234/2018*, in which case the contracted company is jointly and severally liable with the contracting company for the technical and operational aspects of the inherent activity ***(20)***.

GDP is regulated by *RDC n° 430/2020* (modified by *RDC n° 653/2022)* which requires suitable and adequate procedures, premises, installations, and equipment that will ensure proper storage and distribution of medicinal products. This requirement includes temperature-mapping of distribution routes, temperature-monitoring during product distribution, and shipping qualification for all medicines (not only temperature-sensitive products) employing a risk-based approach for product quality ***(21, 22)***. Humidity monitoring during the last mile is not considered necessary as properly chosen container closure systems protect against any humidity excursions which might be experienced. However, Brazilian regulation does require humidity monitoring unless a risk assessment conducted by manufactures and wholesalers determined there is no product impact prior to last mile. Overall, product distribution within the temperature and humidity approved in the regulatory application is an expectation, and the range of allowed temperature and humidity excursions must be technically justified, supported by data, and documented in a risk analysis.

The MAH of a medicinal product must ensure appropriate and continued supply of that medicinal product. In event of a temporary supply discontinuation, a notification to health authorities is required, according to the *RDC n° 18/2014* ***(23)***.

The main challenge in last-mile transport via road, air, and water, is the size of Brazil as the mode of transport is executed by a logistic operator who subcontracts trucks, flights, and boats.



Note: Private supply in red, public supply chain in green *** Sistema Unico de Saude (Unified Health System)**

Figure 3.5-1 Example of Brazil's Supply Chain

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3.6 Supply Chains in the United States

In the United States, there is little difference between the public and private supply chains as public-health efforts normally use private supply chains to achieve their goals. Most prescription drugs travel through a complex system that includes large primary wholesalers, smaller secondary wholesalers, and a small number of specialty wholesalers. In addition to purchasing overstock from manufacturers and primary wholesalers, secondary wholesalers buy and sell drugs among themselves, as well as selling them to pharmacies. Thus, prescription drug products may undergo multiple handoffs within a supply chain.

In the *Code of Federal Regulations* (CFR), the U.S. Food and Drug Administration (FDA) defines wholesale distribution as the "distribution of prescription drugs to persons other than a consumer or patient…." (Part 205.3 (f)) and defines a wholesale distributor as "anyone engaged in wholesale distribution of prescription drugs…." (Part 205.3 (g)) ***(24)***. Additionally, the CFR defines an authorized distributor as "a distributor with whom a manufacturer has established an ongoing relationship to distribute such manufacturer's products" (Part 203.3 (b)) ***(25)***. Wholesalers must have a method to ensure that shipments of prescription drugs are only sold and delivered to licensed entities.

Neither *21 CFR 203* nor *21 CFR 205* addresses packaging and shipping requirements for temperature-controlled products. However, *U.S. Pharmacopeia (USP*) General Chapter <659> Packaging and Storage Requirements and General Chapter <1079> Risk and Mitigation Strategies for the Storage and Transportation of Finished Drug Products are very clear in their guidance for the storage, distribution, monitoring, calibration, training and qualification of "cold chain" equipment ***(26, 27)***. USP <659>, revised in 2020, gives specific guidance on packaging, temperature, storage, and distribution requirements.

Also revised in 2020, USP <1079> provides specific responsibilities, not just for manufacturers but for all last-mile participants, including wholesalers, distributors, pharmacies, compounding pharmacies, hospitals, healthcare providers, brokers, and logistics service providers. It addresses the responsibilities for storage, transport, supplier qualification, quality agreements, excursion handling, training, calibration, monitoring systems, staffing, temperature mapping, and shipping package qualification. USP <1079> focuses on development of a quality management system and includes a section on the "Risk-Based Approach to the Storage and Transportation of Finished Drug Products." Under "Distribution and Shipment of Pharmacopeial Articles," it states that:

> Shippers and distributors are to follow the proper storage and shipping requirements as indicated by the manufacturer….The manufacturers and distributors should work together to establish proper distribution and product-handling requirements for the purpose of ensuring appropriate product maintenance in transit.

In addition, under "Temperature Challenges," USP <1079> states that "Storage temperature ranges may not be indicative of the allowable tolerances during shipping. Articles labeled for special storage conditions (between 2 °C and 8 °C) vary widely in their tolerance of short-term exposure to heat and cold." Manufacturers ship refrigerated pharmaceuticals in a variety of ways, ranging from protective packaging or temperature-controlled trailers to no temperature protection or control at all. The decision about how a manufacturer will ship a product depends on the stability data a manufacturer provides, which may not be included in the product labeling.

Shipments from wholesalers are made via private fleet (vehicles owned by the wholesaler), commercial carriers, and local courier services. Typically, the duration of these shipments from the wholesaler's distribution centers to the point of dispensing facilities ranges from several hours to multiple days.

The National Association of Boards of Pharmacy (NABP) Drug Distributor Accreditation program, (formerly the Verified-Accreditation Wholesale Distributors program) is designed to protect the U.S. prescription drug and medical device supply chain. The accreditation process changed in January 2022

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and no longer requires on-site visits. This program involves a self-conducted supply chain inspection of a facility's procedures and practices. Nearly half of the states require NABP Drug Distributor Accreditation for distributors who are licensed to distribute in their state, whether or not they have facilities in their state, while some states, such as Florida, have a more rigorous licensing process that also involves passing state-certification exams *(28)*. Manufacturers can and often do take state accreditation and licensing into account when evaluating the relationship they have with a wholesale distributor. Other states, such as Georgia and Oklahoma, have started to review their regulatory requirements related to the distribution of finished drug products. Public awareness has also increased in patient claims of improper handling of temperature-sensitive drug products within the last mile that might have adversely impacted patients.

Without specific regulatory requirements, wholesale distributors have only the manufacturer's instructions to ensure they are providing adequate, appropriate, and proper packaging and shipping temperature ranges to maintain product integrity, stability, and potency. In the absence of specific packaging requirements or stability data provided by a manufacturer, the distributor should use qualified packaging or transportation to maintain product-storage labeling requirements for the duration of the shipping time from the removal from the distributor's temperature-controlled environment to delivery at their customer's location.

If available, manufacturers should be encouraged to provide specific packaging requirements or stability data that supports shipping within parameters outside of storage-labeling requirements, since this would reduce environmental packaging waste and costs that are passed on to consumers. In either case, the distributor is responsible for having data showing that the package and shipping method ensures the product's temperature will stay within the storage-label requirement or specified time and temperature parameters. See **Figure 3.6-1** as an example.

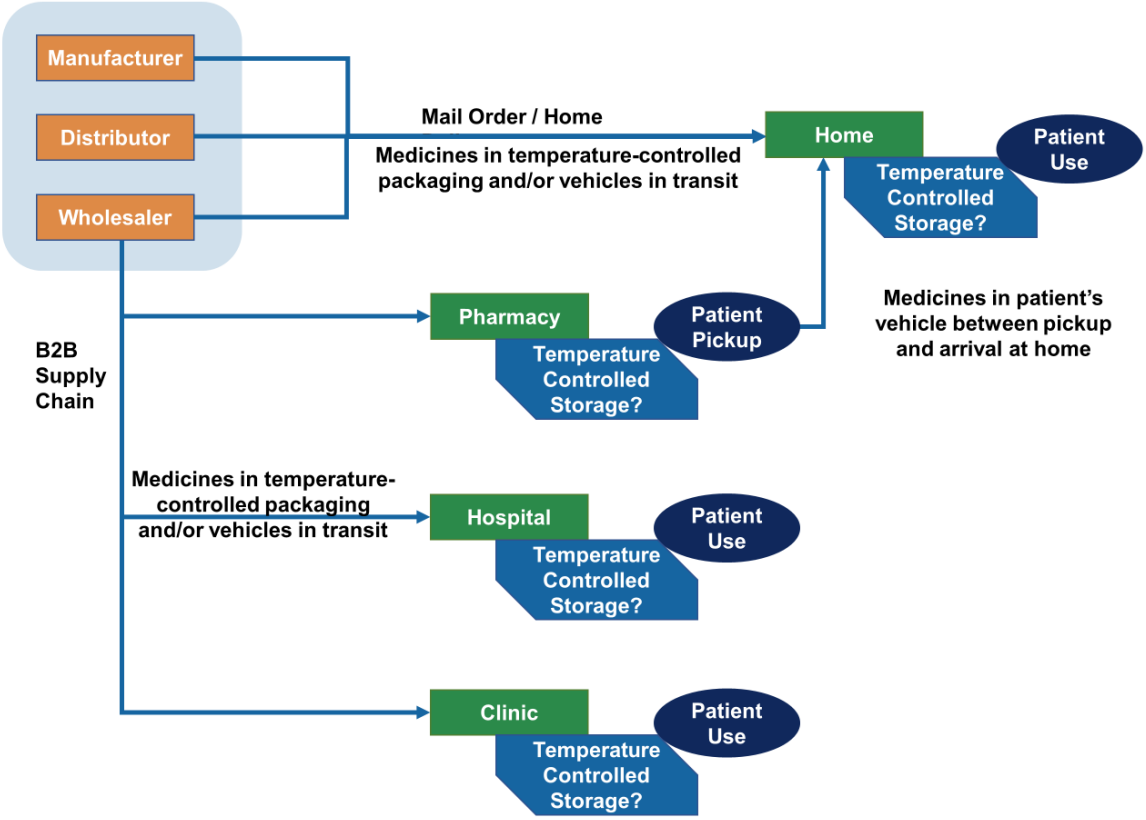

Figure 3.6-1 Example of U.S. Private Sector Supply Chain

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3.7 Supply Chains in Canada

Canada has two main levels of regulating bodies for medicines: Health Canada is the federal regulator, and the Colleges of Pharmacy are the provincial regulators for pharmacies. Health Canada oversees the activities of manufacturing, importation, testing, packaging, labeling, distribution, and wholesaling and, thus, regularly audits and licenses wholesalers.

Each of the ten provinces and three territories in Canada have their respective College of Pharmacy. Hospitals fall under the College of Pharmacy, and they are funded by their respective provincial government or by the government of the territory. Hospitals obtain their licenses from the provincial government where they are established. Pharmacies and clinics, private or public, obtain their licenses from the College of Pharmacy of their respective province.

The Canadian Food and Drug Act (CFDA) and Food and Drug Regulations (FDR) are the two main legal documents that govern the drug, medical device, natural health product, and cosmetics industries *(29, 30)*. The FDR and the *Guidelines for Environmental Control of Drugs during Storage and Transportation* are the Canadian equivalent of the EU GDP*(31)*. Health Canada refers to these documents when auditing any licensed owner of a drug establishment.

Health Canada is also the federal department responsible for monitoring and assessing the wide variety of drugs marketed in in the country. They assess and monitor drugs to ensure that they are in pharmaceutical compliance with the CFDA and its accompanying regulations.

Canada is the second largest country in the world, having an area of almost 10 million km^2, with a relatively small population of 38 million people. Almost 80% of the population is situated in the south of the country, close to the USA–Canada border, which occupies only about 30% of the total area of the country (see **Figure 3.7.1**).

Figure 3.7-1 Map of Canada

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Wholesalers package, pick up, store, transport, and deliver medicines to more than 11,000 pharmacies and hospitals across the country. Transportation of pharmaceutical products can be challenging due to long distances and seasonal temperatures that range from -40 °C to 50 °C, especially when products travel from the south to the north. As a result, products need robust environmental control and protection. In some areas, like in the far north, only one carrier operates and therefore, it is important to assess the route, understand carrier capabilities, and account for the handoff from one carrier to another. It is equally imperative for the wholesaler to have a solid quality agreement with the main carrier responsible for the route in total and with the carrier originally picking up the product at the distribution center.

To preserve drug safety, quality, and efficacy, proper storage and transportation conditions must be maintained throughout the drug supply chain—from the point of manufacture to the delivery of products at the final destination. These conditions are described in the FDR and in the *Guidelines for Environmental Control of Drugs during Storage and Transportation*. Drugs must be stored and transported according to labeled storage conditions or specific transport conditions supported by data. Controls for conditions specified on the label (e.g., temperature, humidity, light) should be in place. Each building used to store drugs must be indicated on the drug establishment license (DEL) warehouse annex. Furthermore, when using contracted third parties to store drugs, it is the wholesaler's responsibility to ensure that the third-party logistics provider's (3PL) buildings are indicated on the DEL warehouse annex, and that the contracted 3PL stores the drugs according to the conditions described on the label.

Packaging and packaging configurations are the responsibility of the DEL owner, who is therefore responsible for designing and validating the packaging used to ship products and for qualifying each route according to a protocol. The executed protocol and the documentation may be presented to Health Canada during an audit.

Considering the difficult Canadian geography, long distances, and extremes in weather, manufacturers are encouraged to share allowable temperature excursions with the wholesaler. Unfortunately, this information is difficult to obtain even when it is part of the quality agreement signed between the wholesaler and the product manufacturer.

3.8 Supply Chains in China

China had approximately 134,000 pharmaceutical distributors in 2021 with a market size of 2 trillion renminbi (RMB), equivalent to approximately 300 billion USD *(32)*. These distributors sell most pharmaceuticals to China hospitals. The Central Pharmacy Department in a hospital is responsible for the purchase and dispensing of pharmaceutical products including inpatient, outpatient, and emergency pharmacies. Although sales are driven to retail pharmacies, especially online pharmacies, hospital sales still account for more than 60% of drug sales. Therefore, the main distribution channel for Chinese wholesalers is the hospital, regardless of whether the medical institution is government- or privately-owned.

A typical manufacturer uses provincial distributors to cover hospital channels, a central purchase department to provide for chain pharmacies and an ecommerce platform to supply directly to the patient. The transaction partners are required to exchange "first transaction documents," including business licenses, drug transaction licenses, and drug sales permits, before a transaction takes place to ensure that the transaction partners have been granted the necessary authorization to deal in a permitted product *(33)*. A distributor is not necessarily bound to limit distribution to within a certain region, but the local administration that issues the distribution license will be held responsible for any conduct contrary to the regulations. Therefore, it is almost impossible for a distributor to open a distribution center in another region, since no authority would issue a license for a distributor not registered in that region.

The pharmaceutical supply chain in China can be divided into the following three steps:

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1. National distribution from manufacturers to wholesalers
2. Local dispatch from wholesalers to hospitals/pharmacies
3. Last-mile distribution from pharmacies to patients

Long-haul distribution is usually carried out with the support of 3PL companies, while the dispatch to hospitals is usually handled by the wholesalers themselves, using their own transportation teams who also serve (partly) as the stocking staff of hospital pharmacies. The transportation team is of crucial importance as the service interface between wholesalers and hospitals. The complexity of such a delivery system makes it almost impossible for manufacturers to replace wholesalers with 3PL companies to provide direct sales to hospitals. Retail pharmacies can dispatch prescription drug (Rx) products in stores or to homes with their own third-party courier services "if the Rx product is to treat chronic diseases and the prescription has been verified by certified pharmacists" *(34)*. Specialty pharmacies, or direct-to-patient pharmacies, serve as the main channel to patients with home delivery services for specialty drugs.

Two critical factors have a great impact on the last mile for pharmaceutical products in China: National Health Insurance and the fast-growing eCommerce channel. The National Health Insurance is the public healthcare insurance system that dominates paying for the healthcare of Chinese citizens. Apart from reimbursement pricing, volume-based purchasing, and diagnosis-related groups, the National Health Insurance has one determinant for expense control—budgeted expense for providers. The management of the budget is also related to the key performance indicators for the hospital management team. With this philosophy in mind, not all retail pharmacies enjoy "reimbursement status" for their patients, even for chronic prescription products. The second factor of eCommerce channels involves online pharmacies selling prescription products with the support of telemedicine offered by the platforms. Such purchases are not reimbursable, but they are convenient in terms of access and delivery.

According to Menet (menet.com.cn), a highly recognized data source for the healthcare market, online sales for drugs grew by 37.6% in 2022 over 2021, with a seven-year compound annual growth rate (CAGR) of 48.4%, accounting for 2.8% of the total drug sales. Concurrent to the development of eCommerce is the emergence of online-to-office (O2O) platforms. O2O is a phrase used in digital marketing to describe systems that entice consumers in a digital environment to make purchases of goods or services from physical businesses. These O2O platforms use carriers that link the stores and customers, and they offer access to retail pharmacies so users/patients can select a product from the listed retail pharmacies according to preferred brand, price, and distance, for example, have it delivered within 30 minutes. O2O business has become an important online outlet for retail pharmacies with thin margins. Healthcare providers in China are mainly state-owned. Grade-3 hospitals, predominantly medical university-affiliated hospitals, are also state-owned, but private providers are gaining in growth rate. Further information can be found in the China Statistical Yearbook 2022 (https://www.stats.gov.cn/sj/ndsj/2022/indexeh.htm). In 2021, the top 10 wholesalers, with a total revenue of 1.3 trillion RMB (about 194 billion USD), accounted for 65% of the market; the top three of those account for 68.35% of the top-10 sales with revenue of 896 billion RMB (about 133 billion USD). The top performers in distribution are also publicly listed companies, the majority being state-owned. To the contrary, the majority shareholders of the leading retail chain pharmacies are private entrepreneurs, although the chain pharmacies are mostly publicly listed companies.

In short, the supply chain for pharmaceutical products in China is shifting from hospitals to retail pharmacies, and online pharmacies are growing at a faster rate than offline retail pharmacies. Ownership does not affect the nature of the healthcare supply chain, but the nature of healthcare products might.

In summary, the pharmaceutical supply chain and last-mile delivery of pharmaceutical products can vary significantly from region to region, with each region presenting different challenges and

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considerations. It is important for companies involved in the distribution of pharmaceutical products to understand the local regulations and requirements to ensure a safe and efficient delivery process.

4.0 Pharmaceutical Products

The pharmaceutical supply chain distributes different kinds of products to the patient via the last mile. Product characteristics and stability impact how these products should be handled along the chain and, therefore, understanding of these product properties is important. The product characteristics, stability, and requirements are further explored in **Sections 4.1–4.2**.

4.1 Product Characteristics and Stability

The chemical and physical properties of a drug substance and the drug product formulation determine its product stability and shelf life *(35-37)*. During this shelf life, pharmaceutical products like vaccines, tablets, creams, and injectables are exposed to hazards that may impact their quality, efficacy, safety, and aesthetics. These hazards, including intrinsic factors such as the molecular structure of the drug itself and environmental factors (e.g., temperature, pH, buffer species, ionic strength, light, oxygen, moisture, additives, excipients), may affect the rates of chemical and physical degradation of drug substances *(38)*.

Temperature is one of the primary hazards that effects drug stability and shelf life. Thus, storage and shipping temperatures should stay within the allowed and filed conditions established during stability-testing programs that are typically run at the beginning of the product lifecycle. These studies determine the impact of temperature, moisture (relative humidity), light, shock, vibration, drop, and pressure on the product in its primary marketed packaging. Here the formation of degradation products and decrease in label claim or potency are measured using stability-indicating methods to the end of the product shelf life, which can range from days to years. During these stability studies, the product is stored under strictly controlled conditions in climate chambers at, for example, 5 ± 3 °C and 25 ± 2 °C *(38)*. Samples are drawn from the controlled temperature chambers for testing at regular intervals. From these stability studies, the product shelf life and temperature conditions during manufacture, packaging, distribution, and storage are established *(39, 40)*.

4.2 Application of Product Stability Budget and Mean Kinetic Temperature

Products requiring storage at defined temperatures may experience exposure to temperatures outside of the defined range for a period of time during filling, labeling, inspection, packaging, shipping, and distribution including to the end user. Proper management is essential to achieve an understanding of the potential temperature exposures a product may experience as it is transported to the point of use. Some examples of situations that may arise include:

Transit within a refrigerated truck

Even though a truck has reached the correct set temperature, the temperature may rise temporarily in warm environments or fall in cool environments as the doors are opened for loading and unloading. Temperature-monitoring devices, located on the outside of product containers, will record this change in temperature, and some monitors may exceed the labeled storage claim of the product.

Storing of prescription drugs within a hospital or pharmacy

The refrigeration unit used to store temperature-sensitive pharmaceuticals accidently stops cooling overnight or during the weekend.

Dispensing of prescription drugs within a pharmacy

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A pharmacy will remove a product from the refrigerator to place a patient label on the product and check that the medication is suitable for dispensing (e.g., it has not expired and that the correct product was removed from the inventory).

Shipment of drug product from a pharmacy to an end user

A product requiring storage at a controlled room temperature (USP definition) is mailed to a patient's home and remains in a mailbox or on a doorstep for several hours where it is exposed to ambient summer or winter temperatures.

In the examples above, it would be important to know if the exposures would be problematic to the product. USP <1079> states that deviations or variances in exposure to temperatures outside of the product's label range should be addressed with the manufacturer or MAH to determine if the exposure is acceptable and if the product may be used.

The total allowable time outside the storage condition, based on stability studies, will be defined as the product's time out of refrigeration (TOR) or time out of storage (TOS) and will be used to support excursion temperatures outside the storage conditions. The total TOR/TOS found to be acceptable is divided by site departments to cover the total time required for each stage in manufacturing, shipping, distribution, and use. Any time left over can be used for excursions more than the defined times.

Even though comprehensive stability studies that support handling both within and outside of the manufacturer's control have been executed, the resulting stability data is typically not shared with other parties, including distributors, caregivers, or patients. Typically, product labels and inserts contain only long-term storage conditions, such as, "Store between 2–8 °C (36–46 °F). Do not freeze!" When excursions outside the label storage temperatures do occur, the manufacturer instructs that they should be contacted regarding the ultimate disposition of the drug product. Manufacturers are reluctant to provide expanded standard stability information to other parties, since the same allowable excursions could be used multiple times by those possessing the product without the knowledge of the manufacturer. The cumulative effects of multiple excursions could impact the quality of the product. A better approach would be to use transportation statements in labeling, for example, "Store at 2–8 ºC; excursions to 25 ºC during transport not to exceed 24 hours" or "Store at 15–25 °C; excursions to 2–32 °C during transport not to exceed 24 hours" ***(41)***. Such statements must be supported by stability data and approved by competent health authorities.

Drug manufacturers can facilitate the use of their temperature-controlled products, when product-stability data allow, by providing a wider range of label storage temperatures (i.e., 2–15 °C rather than 2–8 °C), providing separate long-term storage and transport temperatures, providing specific storage and/or handling temperatures for patient use only, and/or by using mean kinetic temperature (MKT).

Using the calculated MKT for a drug product is appropriate to approximate the effects of storage-temperature variations that may occur during product transport in the absence of product-specific kinetic data. MKT calculations for each portion of the distribution process may be used to estimate the impact of temperature excursions on the quality of the drug product ***(27, 42, 43)***.

For brief segments, such as transport, the MKT for the segment is best used in conjunction with upper and lower temperature limits from the manufacturer.

For longer segments, such as storage, calculating the MKT of only the excursion is preferable to using the MKT of the entire segment for evaluating the impact of the excursion.

Within these limitations, MKT and the controlled room temperature and cold temperature as defined by USP are useful in understanding the impact of temperature excursions on product quality over the lifetime of a drug.

Each successive owner of the material—manufacturer, wholesaler, pharmacy, and patient—can, at best, be responsible only for their particular segment, including the storage and transport temperatures and the interpretation of the resulting partial MKT values. Others in the distribution chain will have

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access only to what is provided in the drug's labeling, typically storage statements (e.g., "Store at controlled room temperature," "Excursions permitted 15–30 ºC," "Do not freeze",); therefore, the inclusion of MKT excursion data in the labeling is useful because the manufacturer may be able to use MKT to evaluate the impact of temperature excursions that occur during the last mile, if the patient contacts manufacturer to request help. The manufacturer of a drug product has a unique responsibility, since it is the manufacturer who performs the stability studies to determine the expiration dating of the drug. It is those stability studies, plus any additional studies the manufacturer may choose to do, such as freeze-thaw and temperature-excursion studies beyond the standard ICH data package, that enable informed decisions to be made about the use of MKT to evaluate the impact of temperature excursions on the product. For this reason, when questions arise during the distribution process about temperature excursions, they are best raised with the manufacturer who can use the stability data to interpret the MKT.

At multiple points in the supply chain, excursions outside the labeled long-term storage temperature range may occur. This may be due to a deficiency in the stability information that is necessary for managing temperature excursions experienced throughout the distribution process. The misalignment can be resolved if the logistics group openly communicates the stability needs to the research and development team that executes the stability studies. Comprehensive stability data is essential for managing temperature excursions that are likely to occur during transit ***(36, 41)*.**

5.0 Services and Temperature Controlled Systems

Within supply chain management, suppliers and receivers of goods may use services and temperature-controlled systems from other entities to support the storage and transport of pharmaceutical products. When temperature-sensitive products need to be shipped, additional services may be needed to protect and monitor the goods. **Sections 5.1–5.6** focus on contract management, storage, packaging, and transport for pharmaceutical products.

5.1 Contract Management

When services or outsource activities are applied in the supply chain, contract management should be applied according to the applicable country's rules. The quality system should extend to the control and review of any outsourced activities related to the procurement, holding, supply, or export of medicinal products, and all of these processes should incorporate quality risk management.

In addition, there must be a written contract between the contract-giver and the contract-acceptor that clearly establishes the duties of each party. Besides the commercial contract, from a regulatory perspective, a technical or quality agreement is expected that includes additional details for deviation management and change control. The contract-givers are responsible for auditing the contract-acceptor to verify the agreed-upon activities.

The contract-acceptor should not pass any of the work entrusted to them under the contract to a third party without the contract-giver's prior evaluation, approval, and an audit of the third party by the contract-giver or the contract-acceptor. The contract-acceptor must forward any information that can influence the quality of the product(s) to the contract-giver in accordance with the requirements of the contract.

5.2 Storage

5.2.1 Storage Facilities

Storage facilities should be secure and structurally sound to allow safe storage and handling of the medicinal products. Management and operation of the facility should include but is not limited to:

- Access to the storage facilities should be restricted to authorized staff.

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- Premises should be provided with adequate lighting to enable all operations to be carried out accurately and safely.
- Appropriate cleaning equipment and cleaning products should be chosen and used so as not to present a source of contamination.
- Any products withdrawn, recalled, expired or suspected of falsification should be segregated from saleable stock (either physically or through an equivalent electronic system, when possible).
- Medicinal products should be handled and stored in such a manner as to prevent spillage, breakage, contamination, and mix-ups.
- Stock should be rotated according to the "first expired, first out" (FEFO) principle. Regular stock inventories should be performed, and stock irregularities should be investigated.

5.2.2 Environmental Control

The storage of pharmaceutical products is critical for the last mile knowing the challenging conditions these products will be exposed to until they reach the end users. If applicable, temperature and humidity requirements must be maintained during storage, regardless of the type of storage unit used (e.g., deep freezer, freezer, refrigerator, walk-in cooler or freezer, storage rooms).

To ensure proper storage, three essential conditions must be met: temperature and humidity maintenance, distribution, and monitoring. Temperature and humidity must be maintained during regular operation and should consider traffic movement (door opening) and defrost cycles where applicable. To ensure proper temperature and humidity distribution, storage units should be mapped to ensure that all locations inside do not deviate from the required temperature range (i.e., there should be no exception, even for small units like an under-counter refrigerator). The temperature- and humidity-mapping, in regions where this is necessary, should be conducted when the unit is empty *(44)*. However, additional studies might be required based on the geographic location, atmosphere, and local regulations (e.g., winter/summer season, dry/wet seasons) in combination with partial or full loads, as product placement may affect air movement. Once temperature-mapping is executed, the next step is analyzing the thermal mapping data to identify the locations with the lowest and highest values where temperature and humidity sensors can be positioned for permanent monitoring.

In some cases, temperature probes are placed inside a container with liquid to provide a more accurate product temperature rather than air temperature, which can fluctuate and trigger an alarm unnecessarily. Many electronic sensors can simulate probes in liquid to alleviate temperature changes and reduce nuisance alarms. A contingency plan must be in place in the event of a power outage or mechanical failure. Standup freezers and refrigerators can only maintain the proper temperature for a few hours during failures and are not typically designed with significant insulation properties (particularly models with glass doors).

The medication provider (e.g., pharmacist) must communicate to end users the importance of proper storage conditions and provide guidance on the best ways to store the product prior to use. If the product requires refrigeration, the products will be stored in a refrigerator. Refrigerators used at a home often cool the interior space with cold air from the freezer section of the unit. Placing the products near the vents supplying cold air may expose products to freezing temperatures. The same situation can happen when a refrigerator is over-packed and prevents air from circulating correctly. Along with requirements to protect the product from excess heat and/or freezing, the impact of light and humidity can be substantial and should be explained to end users, if applicable.

5.3 Packaging

The diversity of environmental conditions encountered in the last mile makes this segment the most challenging of the whole distribution chain. For this reason, the qualification of packaging requires a deep analysis of potential environmental and handling conditions that will be experienced until the product is delivered to end users. PDA *Technical Report No. 64: Active Temperature-Controlled Systems:*

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Qualification Guidance, *Technical Report No. 72: Passive Thermal Protection Systems for Global Distribution: Qualification and Operational Guidance*, and TR 39 detail all the processes for qualifying a thermal solution *(1, 44, 45)*. The most crucial step in this process, however, is to design proper testing conditions before starting the qualification process. Any scenarios that the products will encounter must be taken into consideration (e.g., design qualification, operational qualification). For instance, if the product will be delivered by car or truck, with or without active heating and cooling, or with couriers or parcels, the conditions inside the packaging, the number of stops, and how the products are tendered to the end users need to be known in the development of testing temperature profiles and handling procedures. Data collection is crucial in the establishment of qualification procedures. Using dummy boxes to collect outside temperatures and measure handling conditions (e.g., vibration, orientation) during shipments is a very effective method for understanding the shipping environment that will be used for qualification. In the case of parcel shipments, thermal shipping containers can be flipped multiple times, increasing the probability of a package landing on its side.

The role of packaging in the last mile is fundamental. Earlier in the supply chain, most pharmaceutical products would have been transported in larger quantities (e.g., pallet size, multiple-counts package). For the last mile, products are shipped in much smaller quantities for the final deliveries such as those to pharmacies, clinics, nursing homes and, eventually, directly to patients. These shipments will experience more diversified conditions than they had in their homogeneous distribution. Considering the time and conditions under which the packaged products stay at their origin until they have been picked up for transport and delivery is also important. In some cases, and in some regions of the world, packages are stacked on a pallet, rolling carts, or shelves for several hours before being picked up. These conditions need to be considered during design and qualification. In addition, sustainability (e.g., single use versus reuse), theft, and all means of potential damage should also be factored into the packaging design.

5.3.1 Packaging with Passive Thermal Functionality

Packaging with passive thermal functionality uses a combination of insulation and refrigerants, such as water-based gel packs, phase-change materials (PCM), dry ice, or liquid nitrogen to ensure that medicinal products are distributed under defined temperature conditions. Regardless of the options selected, containers must be qualified *(46, 47)*. A prequalified passive-thermal solution can be acceptable as long as the ambient temperature for which it was qualified and the payload used for the qualification is similar to the product to be shipped. The vendor qualification documentation of the prequalified solution must be reviewed and assessed by the shipper to confirm its suitability for the intended use. Universal pack-outs do not require seasonal or regional adjustments and maximize operational efficiency but usually are more expensive and weigh more, resulting in a higher logistics expense. Overall, the product mass, container-insulation thickness, performance hours, and thermal stress will affect the size of the required passive-thermal solution with refrigerants and thermal barriers, which in turn factors into material costs, warehouse space, and carrier charges. Substitutions of any components should be tested to confirm that the same thermal performance is achieved. Live shipment testing is strongly recommended to confirm the performance capability of the designed thermal shipping system and process is effective and reproducible in representative real-world distribution lanes with actual transportation process and logistics service providers *(47)*. Factors related to home deliveries that include additional transport hours and time packages are left outside or in mailboxes must be considered in the final design of the thermal-packaging solution.

5.3.2 Packaging with Active Thermal Functionality

Several different systems with active thermal packaging are available which are equipped with active refrigeration or heating capabilities using calibrated and temperature-control devices. The refrigeration or heating used to keep the products under proper temperature conditions can be mechanical, chemical, or electric and typically is designed to be reused multiple times. Contrary to previous segments of the supply chain, active-thermal containers are much smaller and accommodate a small

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number of products. Most often, their size is that of a small portable cooler used mainly for delivering products to nursing homes, hospitals, clinics, or vaccination centers. Usually, these systems can only maintain a temperature of 10 °C to 20 °C below the ambient temperature. Like passive-thermal packages, active-thermal containers must follow the same process for qualification. Since these systems are designed to be used several times, an inspection process must be in place to ensure that they will perform the same each time.

5.3.3 Sustainability in Packaging

Pressure to be more sustainable has reached the last-mile segment. More countries and users are asking for greater environmental consideration in the usage of packaging for the last mile. Much passive-thermal packaging is single-use and made of materials that can be difficult to recycle and are not biodegradable, including expanded polystyrene (EPS) and polyurethane. New materials are available that provide a more sustainable solution such as starch-based, paper, cotton, or wood-fiber packaging that is recyclable, biodegradable, or compostable. There are limitations to many of the new sustainable materials, however. Some are susceptible to water damage and may require the use of plastic wrapping. Additionally, some of the materials may have lower thermal properties compared to traditional materials and require thicker insulation or additional refrigerants to provide comparable thermal performance. Finally, the cost for these new sustainable solutions can be higher than traditional insulation materials.

Another strategy to improve sustainability is to employ reusable solutions. Last-mile delivery provides an opportunity for reusable packages when end users can easily and cost-effectively return thermal solutions. Sustainability related to reusable solutions can sometimes be complex, as sometimes the carbon footprint of reverse logistics can be greater than a single-use solution with sustainable materials. When using a reusable solution, a documented assessment procedure for each solution must be done to ensure their performance in the last mile. For example, it is essential for packages made of vacuum insulation panels (VIP) to be inspected before use as the loss of the vacuum will significantly diminish the thermal performance of the shipping solution. Also, before reusing thermal solutions, sanitation procedures could be required.

5.4 Transportation

Various transportation methods are used within the last mile (e.g., air, truck, car, bike). Regardless of the method, temperature should be maintained to the label claim on the product's outer packaging or the manufacturer's instructions. Most transport within the last mile is national and each carrier may have different transportation methods. Wholesale distributors and local couriers, for example, typically use ground methods of transportation. Understanding the method of transportation is important to ensuring that proper storage conditions can be maintained throughout the delivery process. Considerations for any appropriate transportation method should include the staging process, weather, length of transit, and accessibility to the patient; likewise, qualification testing should include each of these considerations. It may be necessary to stop during certain weather conditions, e.g., certain months of the year, such as the rain season in Africa and Asia or low temperatures in Canada. Similarly, deliveries to countries where certain days are religious holidays or long weekends should be scheduled accordingly. For specific deliveries, even drones may be used. When using specific solutions like drones (e.g., to reach remote patient locations), it must be done in compliance with jurisdictional laws and regulations. Qualification testing for such specific solutions should include any stability issues that may arise from dropping the shipment to the ground.

5.4.1 Road Transportation

Depending on the product, distribution by road might require a temperature-controlled vehicle, like a van, truck, or motortruck with trailer *(48)*. PDA TR 64 defines a temperature-controlled trailer as "a cargo box attached to a truck…that is equipped with a TCU [temperature-controlled unit] to provide active cooling or heating control inside the box…sometimes referred to as 'reefers'" *(44)*. Most

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temperature-controlled trailers are certified under the *Agreement on the International Carriage of Perishable Foodstuffs and on the Special Equipment to be Used for such Carriage* and have one TCU (mono-temp trailer) or have one TCU mounted at the front of the trailer and a second evaporator at the rear doors (bi-temp trailer) ***(49)***. Bi-temp trailers are the most commonly used temperature-controlled road vehicles in domestic and international road transport to keep temperatures at 2 °C to 8 °C, 15 °C to 25 °C, or -15 °C to -25 °C. The bi-temp trailer has a movable split wall that can divide the front from the back of the trailer where the evaporator at the rear doors controls the temperature in the rear section ***(48)***. Bi-temp solutions are also available in the marketplace for vans.

5.4.2 Drones

Drones may be an interesting alternative transport mode to deliver low-volume product(s) quickly in 5 to 60 minutes over a distance up to 100 km to remote/rural areas, such as coastal islands, mountain areas, tropical forests, areas with poor infrastructure and infrequent logistics, and/or in case of emergency. Delivery drones can be piloted manually, highly automated, or fully automated under supervision of a qualified pilot, but will not take off in severe weather conditions (e.g., hail, snow, heavy rain, wind). Drone delivery recipients can be trained remotely in less than ten minutes to be familiar with drone unloading practices and/or refueling (e.g., exchange of batteries). In the near-term, drone transportation is relatively expensive, so routine drone flights would mainly be an option for high-value products.

While drone transport is by definition air transport regulated by national and/or international aviation authorities, there are key differences versus shipping using conventional aircraft. A package sent via drone is exposed to ambient conditions in a way that a package in the cargo hold of an aircraft is not. For this reason, it is essential that the packaging provides any necessary temperature and weather protection. In addition, drone transportation has the potential to expose the contents to shock, drop, vibration and numerous other factors that could impact the product in a manner that may be very different from that experienced in the cargo hold of an aircraft. On the other hand, drones fly up to 120 meters (about 400 feet) above the ground, which is not high enough to create the pressure concerns experienced with aircraft. As the use of drones for the last mile increases, it will be important to provide appropriately protective secondary and tertiary packaging for pharmaceuticals and vaccines transported via this mode. Specific protection should be considered based on the type of risk and qualification if the critical process parameters are met.

5.4.3 Distribution from Point of Dispensing to End-Use in Mail Service Pharmacy

The traditional dispensing role of pharmacists has shifted to include a clinical focus in monitoring the appropriateness of prescribing and in-patient care. Automated programs, including mail-service pharmacies, allow the time for these additional clinical responsibilities. Given the shift in dispensing trends, quality programs used throughout the pharmaceutical industry to transport medications must also be implemented at mail-order pharmacy facilities as well as at local pharmacies.

When mailing pharmaceuticals for home delivery, temperature assurance should be maintained until the patient can return the product to its proper storage environment. When selecting the appropriate transport company, thermal solution, or theft protection, mail-order operations must especially focus on thermal conditions. The duration for the thermal qualification should start at the time the first package is packed until the end user opens it. For example, next-day shipping should have a qualified thermal solution for 36 hours that considers the time the first package is packed, delivered to the end user's address (doorstep or mailbox), and opened by the patient at the end of the day (after working hours). This example, however, does not consider all the issues involved when the time frame of delivery is calculated; in many cases, significantly more time may be needed.

Due to the expansion of mail-order pharmacies, many facilities must ship to multiple regions, which increases the complexity of selecting the appropriate thermal solutions. Many pharmacies currently

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choose thermal solutions using a dynamic weather system based on the weather conditions of the origin and final destination. Dynamic weather systems provide a more accurate estimate of what the package will experience but also require having multiple qualified thermal solutions for different weather scenarios, such as cold to warm and warm to cold. Pharmacies that do not use dynamic weather systems most often use a universal thermal solution.

Strategies to improve delivery include but are not limited to:

- Minimize transport hours
- Decrease the time packages are left outside, in a mailbox, or on a carrier's truck
- Use prequalified or validated passive-thermal solutions or active temperature control (e.g., reefer unit)
- Include temperature indicators or loggers on the product or packaging
- Develop and maintain ongoing quality programs

5.5 Temperature-Monitoring During Transport

Global regulations for the transport of temperature-sensitive pharmaceutical products specify that transportation is an extension of storage. Therefore, temperature should be maintained to the label claim on the product's outer packaging or the manufacturer's instructions during transportation throughout the entire supply chain, including the last mile.

To achieve this, the shipper may use qualified temperature-controlled shipping systems such as active- and/or passive-thermal solutions. Although these solutions may have been qualified to provide assurance that they will provide adequate thermal protection during shipping, unforeseen events may cause temperature excursions. The only way to provide absolute assurance that the required temperature range has been maintained is to include temperature monitors, data loggers, or indicators.

5.5.1 Temperature-Monitoring or Data Loggers

A wide variety of temperature-monitoring or data-logging solutions are available for the tracking of temperature during transportation. One option is "passive" USB data loggers which can be programmed with predefined settings (e.g., logging of internal temperature, a start-delay, alarm limits), started, and applied to the shipment. Upon receipt of the shipment, the data logger is retrieved, and the data downloaded and reviewed to support the decision about product disposition. There are also solutions that use near-field communication or Bluetooth, allowing for easier retrieval of the data upon receipt of the shipment.

Near-real-time ("active") data loggers are also available which use global positioning systems and cellular communication networks to transmit data "live." They allow the shipper to view the temperature data mid-shipment, with the possibility of intervention should an event/issue occur which could affect product quality.

Temperature data loggers (passive and active) provide a full set of temperature data for the shipment, including a temperature reading or recording at a set recording interval (e.g., every 10 minutes) as well as a detailed statistical analysis of the shipment (e.g., cumulative time outside a defined limit). Even in the event of an alarm triggered by the data logger, an analysis of the full data set may still allow the product to be released if the temperature excursion is found to be within the allowable limits based on stability data for the product.

5.5.2 Temperature Indicators

Temperature indicators come in the form of chemical or electronic indicators. These are a lower-cost alternative to data loggers but, whereas a data logger provides a full set of temperature data for shipment, including statistical analysis, an indicator will only confirm if the required temperature range

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was maintained. Temperature indicators do not generally show when the temperature excursion occurred, however, some electronic indicators will allow a cumulative alarm to be programmed into the indicator.

5.5.3 Risk-Based Approach: Implementation and Management

Implementing temperature-monitoring in transportation across the end-to-end supply-chain is highly recommended, however, it may not always be possible to use temperature-monitoring solutions within the last mile. A risk-based approach should be used to assess the suitability of using temperature-monitoring solutions *(50, 51)*. Where it is assessed as not suitable or practical (e.g., shipping directly to a patient who may struggle with the technology), time-based qualified shipping solutions may be implemented.

As well as using an independent temperature-monitoring solution, if the product is being transported in an active thermal solution, it may be appropriate to use the vehicle temperature-monitoring system. Whatever system is used, it is vital to make sure the device is properly calibrated to ensure that the recorded data is accurate. The precision of the device should be at least ±0.5 °C.

The location of the temperature-monitoring solution is also very important. The location should be based on the temperature-mapping or qualification of the shipping system and placed to represent the product temperature as accurately as possible. Poor placement could result in a true temperature excursion being missed and product being released when it should be discarded, or a temperature excursion being recorded due to measurement of air temperature when, in fact, the product quality is not affected. Best practice is to put the temperature sensor next to the product(s) shipped.

Using reputable providers and solutions, executing an appropriate assessment and/or qualification, and assuring a robust implementation and ongoing management process will ensure product safety and efficacy.

5.6 Security

Security measures taken during the last mile are intended to ensure only the intended recipient receives the package. This may include signature-required, tamper-proof, and anonymous packaging. The cost of using these methods is balanced against the risk mitigation for each medication, transportation method, and consumer. The shipper needs to analyze the use of security methods consistently.

6.0 Data and Knowledge Management

Data is critical to manage and prove that products are delivered with the right quantity and quality at the right time and under the right conditions. Data can be product information, transport and storage conditions, track-and-trace or traceability data, serialization numbers, and other information such as supply-chain disruptions related to traffic jams, weather conditions, and inventory levels. Data can be static or dynamic. Static data does not change after being recorded and is a fixed data set, whereas dynamic data may change after it is recorded and must be continually updated (e.g., temperature data, location track-and-trace data of shipments). **Sections 6.1–6.1.3** describe the management of data in relation to GDP and good documentation practices (GDocP), including the ALCOA+ data-integrity principles.

6.1 Good Documentation Practices and Data Integrity

Since 2015, regulatory bodies like the EMA, FDA, WHO, and Medicines & Healthcare products Regulatory Agency (MHRA) have had an enhanced interest in ensuring data integrity. Each has published data integrity guides to promote the transformation of the pharmaceutical industry to an elevated data integrity level. Initially, the main focus was on testing laboratories and manufacturing,

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but the guidelines also currently apply to distribution to ensure the quality, efficacy, and safety of the product is maintained according to local requirements *(3, 9, 11, 52)*.

6.1.1 Data Governance

Data governance encompasses the people, processes, and technology arrangements required to achieve data integrity. Assuring data integrity requires appropriate quality and risk management systems, including adherence to GDocP and local regulations, such as General Data Protection Regulation (GDPR) in Europe, that:

- Address data ownership, access, and accountability for processes and risk management throughout the entire data lifecycle
- Consider an adequate design, operation, and monitoring of processes and systems
- Include controls over intentional and unintentional breaches of data integrity

Data-governance design and controls must enable the detection of errors, lapses, and omissions of results and data during the data lifecycle. Controls can include procedural, organizational, and functional (technical) controls.

The data governance system must include enough detail to allow accurate and complete reporting and interpretation of all data and must ensure data integrity. This system should include, but is not limited to, the following elements:

- Validated computerized systems that are fit for their intended use, with special attention to any system used to create, process, and store data
- Systems and procedures in place to ensure that records are reliable, complete, and accurate; these systems and procedures must also require that all test results that could affect the quality, safety, or efficacy of a drug are reported, reviewed, and assessed appropriately
- Organization and storage of data in a way that is interpretable and traceable to the execution and purpose of distribution procedures
- Controls in place to ensure that test data are not deleted and that changes to records are documented and justified, where required (e.g., audit trails should be enabled and reviewed)
- Data retained in its original format; original records (or true copies), including electronic records, are subject to review by qualified personnel

Risk management is an essential component of the data governance system. A risk-based approach to data management ensures that adequate resources are allocated and that control strategies for the assurance of the data integrity are commensurate with their potential impact on patient safety, product quality, and compliance with regulations. Where risks are identified, appropriate measures or safeguards should be implemented to ensure the integrity of critical data and records.

6.1.2 Good Documentation Practices

According to the FDA and other inspectorates, ensuring data integrity requires that data be attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring, and available (ALCOA+). These general principles of GDocP apply to the management of records throughout its lifecycle, regardless of the medium (i.e., paper or electronic records), from the time data is first generated and any modifications made thereafter *(52)*.

The EU GDP provides guidance in Section 3.3.1 on how to manage data *(53)*:

- Data should only be entered into the computerised system or amended by persons authorised to do so.

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- Data should be secured by physical or electronic means and protected against accidental or unauthorised modifications.
- Stored data should be checked periodically for accessibility.
- Data should be protected by backing up at regular intervals. Back-up data should be retained for the period stated in national legislation but at least five years at a separate and secure location.
- Procedures to be followed if the system fails or breaks down should be defined. This should include systems for the restoration of data.

The WHO guidance differs slightly from the EU GDP in the retention of the data, indicating that "all monitoring records should be kept for a minimum of the shelf-life of the product distributed plus one year, or as required by the national legislation" *(11)*. However, national legislation may differ from either of the WHO or EU GDP guidances; for example the United Kingdom requires a two-year retention period, after which time records can be destroyed.

Each paper and/or electronic record should be secured against modifications and/or deletions to the end of the retention time as defined by the applicable regulations. Therefore, a paper record should be traceable and stored in a secured environment (e.g., locked, fire-proof cabinet), whereas an electronic record should be secured in a computerized system which employs system access (segregation of duties), vulnerability scanning, security firewalls, antivirus applications, and security patches. In addition, risk assessment of the computerized system should provide requirements toward data business continuity management and recovery plan, data integrity, system weaknesses against human interactions, data manipulation, confidentiality, data privacy, and so on. In the procurement of new computer systems, consideration should be given to security, audit trails, data transfer, and data backup capability *(53, 54)*. Using standalone systems should be avoided or mitigated where possible. Data management also requires a validation that secures the data, including training, risk assessment for outsourced activities (if applicable) such as use of cloud computing, system maintenance, backup system features, and data retirement. Cloud computing services are to be treated as a contracted service, with the risks identified and addressed, and the responsibilities of both parties clearly outlined in a suitable contract or service agreement. If storing electronic records in a PDF format, a risk assessment should be completed on the editability of the document. Some security settings for PDFs allow editing, which would not comply with requirements applicable to manufacturers in *21 CFR Part 11: Electronic Records; Electronic Signatures—Scope and Application* and EU GDP *Annex 11: Computerised Systems*.

6.1.3 Data Integrity and Quality Management

There should be written procedure for a data-integrity and data-security assessment that identifies data integrity and data security risks and ensures that suitable procedural or technical controls have been applied. Any detected or potential data integrity or security breach must be documented as a deviation and investigated through the identification of the root cause and the determination of the resulting corrective and preventive action (CAPA) plan. Some examples of data integrity breaches include:

- Missing or untraceable data entries or archived paper documents
- Missing a proof of delivery document
- Missing temperature data logger
- Data accidently or intentionally omitted or destroyed
- Deficient electronic data backup or archive
- Incorrect calculations impacting decisions
- Unjustified alterations or manipulation of data (e.g., backdating)
- Wrong access role used (e.g., administrator role used for routine tasks)

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- Data review not completed on a signed off report
- Failure to comply with the ALCOA+ principles
- Loss of data during data transfer or due to system failure
- Sharing of user accounts or passwords

If the data-integrity breach may have an impact on product quality and/or patient safety, the violation must be escalated to management and communicated to customers (as applicable). An assessment of the breach must be conducted and the relevant regulatory or health authorities (e.g., country, local) must be notified, if required by local regulations. More information can be found in the FDA *21 CFR Part 11* and EU *GMP Annex 11*.

6.2 Temperature Management

Temperature is a critical process parameter that must be monitored and controlled in the supply chain during manufacturing, storage, and transportation to ensure that pharmaceutical products comply with the critical quality attributes to the end of their shelf life. To secure temperature control of pharmaceutical products in the supply chain, five basic principles should be applied by each supply-chain stakeholder *(41, 55)*:

1. Temperature specifications are set for products, processes, and systems.
2. Temperature is controlled in an environment within the intended range using qualified active and/or passive system(s).
3. Temperature is measured by calibrated sensors at hot or cold spot location(s) based on temperature-mapping studies.
4. Temperature data is collected and securely stored in readable and retrievable electronic and/or paper format according to ALCOA+ data integrity principles *(53, 54)*.
5. Temperature excursions are investigated and product dispositions are executed according to current GDP *(9, 51)* and good pharmacy practices *(56)*.

In event of temperature excursions, the product impact assessment and product disposition should be based on the generated data. However, additional non-GxP data (metadata) can be used by the business to gain a better understanding of the circumstances of the temperature excursion and to determine CAPAs for continuous improvement. In the end, temperature data storage, whether on paper or on a third-party client's cloud solution, needs to comply with all relevant regulatory requirements for security and data integrity. This will ensure that all collected and recorded data is complete, consistent, and accurate throughout the data lifecycle.

6.3 Other Critical Process Parameters

In addition to verifying the shipment and receipt of products, monitoring devices may record temperature at other points in the supply chain. Other possible enhancements that may prove useful include detecting the opening and closing of packing containers and providing a geofence. A geofence is a virtual perimeter for a real-world geographic area; it could be dynamically generated or matched to a predefined set of boundaries. The degree to which these devices may replace or supplement temperature monitors and recorders remains to be determined.

In the last mile, finished products are protected against humidity by the primary packaging and against light by the primary/secondary packaging. Therefore, removal of secondary packaging is not recommended as light may impact product quality, safety, and effectiveness. If a product is moved

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from a hot, humid environment to a cold environment, that product may get wet due to condensation which may impact readability of a batch number or expiry date.

In the case of a mail-order pharmacy, most shipments use water-based cool bags, packs, or bricks (refrigerated or frozen) in "thermal pack-outs." These bags may leak on occasion due to initial physical damage done while frozen (e.g., dropped, impacted); these leaks will generate a significant amount of water in the secondary packaging. It is important to use a sealed bag to protect both the pharmaceutical product as well as the pharmacy prescription label. The use of dunnage is also advised to prevent movement inside the outer package. Air pillows are not recommended because their volume will decrease significantly at lower temperature and will not provide significant protection.

6.4 Serialization

Serialization is the assigning of a unique serial number to each saleable unit of every prescription product. It provides the product's origin, and other information in the form of a barcode or two-dimensional (2D) data matrix. This also serves as a unit's tracking or traceability system through the entire supply chain–from production, through distribution, to the point of dispensing to the patient.

Serialization of pharmaceutical products has been introduced in recent years in many countries to protect the patient against counterfeit and falsified drugs. Counterfeit drugs can prove ineffective, harmful and, in some cases, fatal. Falsified medicines are often disguised as authentic medicines but may contain ingredients of poor or toxic quality or in the wrong dosage.

6.4.1 Serialization in the European Union

To address the threat of falsified medicines in the EU, Directive 2011/62/EU and its Delegated Regulation 2016/161 include measures aimed at preventing the entry of falsified medicinal products into the legal supply chain. The Directive introduces harmonized EU measures to fight medicine falsifications; it ensures that medicines are safe and that the trade in medicines is rigorously controlled including *(57, 58)*:

- Obligatory safety features–a unique identifier and an antitampering device on the outer packaging of all prescription medicines
- A common, EU-wide logo to identify legal online pharmacies
- Tougher rules on the import of active pharmaceutical ingredients
- Strengthened record-keeping requirements for wholesale distributors

Since 2019, the EU has required manufacturers of prescription medicines to apply a 2D barcode for the serial numbers, which must be unique and must be randomly generated. In addition to the serial number, the data matrix code contains the product code, batch number, expiration date, and (if necessary) national reimbursement number.

Once the manufacturer has sealed the medicine boxes with a tamper-evident seal and affixed the data matrix, the serial numbers are uploaded in the central database (EU hub) and downloaded to the databases of the member state where the boxes are being sent. It is possible for a consignment of medicines to go to another country, but that triggers an Intermarket Transaction. If the target market of a medicine changes, the batch's serial number is sent via the EU hub to the database of the correct country.

Directive 2011/62/EU obliges pharmacists and healthcare professionals who supply medicines to the public to check that the tamper-evident seal of the medicine packages was not tampered with and, then to deregister the unique serial number on the medicine packaging before dispensing it to a patient *(57)*. In addition, the unique serial numbers in the EU hub can be traced to all national medicines repository systems in which the package could be dispensed.

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Wholesale distributors must verify all packs of medicines they buy from other wholesalers as well as any packs returned to them. They must decommission products they export outside the EU and at the request of member states for other entities (Article 23) who are not themselves connected to the national medicines verification system in their country.

The pharmacist decommissions each pack in the central database by scanning the 2D code. If the pharmacist receives an alert indicating that the serial number does not exist or has already been decommissioned, or if the tamper-evident seal has been tampered with, the pack must be quarantined by the pharmacist until the case is resolved. If a potentially falsified medicine is identified, the pharmacist must report it to the National Competent Authority. A written, approved procedure should be available and followed to deal with such cases.

6.4.2 Serialization in the United States

Section 913 of the Food and Drug Administration Amendments Act of 2007 (FDAAA; Public Law 110-85) amended the Federal Food, Drug, and Cosmetic Act to create "Sec. 505D. Pharmaceutical Security." This section requires the U.S. Secretary of Health and Human Services to develop standards as well as "identify and validate effective technologies for the purpose of securing the drug supply chain against counterfeit, diverted, subpotent, substandard, adulterated, misbranded, or expired drugs" *(59)*.

In March 2010, FDA published the *Guidance for Industry: Standards for Securing the Drug Supply Chain - Standardized Numerical Identification for Prescription Drug Packages* for identifying package-level standardized numerical identifiers (SNI) as an initial step toward facilitating other measures for securing the drug supply chain. Although the FDA did not specify a particular means of incorporating SNIs onto the unit package at that time, the SNI was compatible with existing machine-readable forms of data carriers, such as the 2D matrix barcode and radio frequency identification (RFID).

While RFIDs and barcodes can carry the same product identification and serial number data, the RFID has the advantage of being read remotely. Regardless of the data carrier, when the product ID and serial number are linked, the supply chain will be further secured, down to the last handoff to the patient through authentication of the product and its pedigree via a database that contains the appropriate manufacturer, distributor, and pharmacy transaction data.

While the 2007 FDAA, 2008 H.R. 5839, and the 2010 FDA guidance were initiated separately, they led to a convergence within FDA on using electronic track-and-trace technology with serialization down to the unit-of-sale level for prescription drugs. And in 2013, to decrease the production of counterfeit, stolen, contaminated, or harmful drugs in the United States, Congress signed into law the Drug Quality and Security Act (DQSA).

The Drug Supply Chain Security Act (DSCSA), Title II of the DQSA, requires that every prescription drug package and homogeneous case have its unique identifier in both machine- and human-readable forms. The Act outlines steps to achieve interoperable and electronic tracing; this means that, in addition to label serialization, a data system would need to store product traceability information showing the chain of custody. These regulations require all trading partners in the pharmaceutical supply chain—manufacturers, wholesalers, and dispensers—to use secure, interoperable electronic systems and processes to uniquely serialize each individual drug package in the supply chain and trace their distribution *(60)*.

The Pharmaceutical Distribution Security Alliance (PDSA) is an organization of stakeholders committed to safeguarding patients through better security of the drug supply chain. The PDSA is working with the FDA to develop a single national uniform system for product serialization and traceability with a goal toward unit-level interoperability. Numerous other organizations, such as the Health Distribution Alliance and International Coalition of Medicines Regulatory Authorities, are all working toward similar goals for serialization.

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Pedigree requirements currently vary from state to state and where unit serialization may be used, it is seldom linked to a pedigree. While many pedigrees are paper-based, there are various types of electronic pedigrees currently being used, including the GS1 Drug Pedigree Messaging Standard. GS1 is a leading global organization dedicated to the design and implementation of global standards and solutions to improve the efficiency and visibility of supply-and-demand chains globally and across sectors.

6.5 Technologies

6.5.1 Radio Frequency Identification Tag

RFID refers to an automated data-capture system that uses low-power radio waves to communicate between two components: tags and readers. Readers are devices with antennas that emit radio waves and receive signals back from the tags. Tags use radio waves to communicate their information to readers; they can be passive or active. Passive tags are powered by the reader and do not have a battery whereas active tags are powered by batteries. In the last mile, RFID technologies are used for inventory control, tracking and tracing products, verification of shipments and receipt of products, or finished product authentication. RFID tags may also record temperature within the supply chain or detect the opening and closing of packing containers. The degree to which these devices may replace or supplement temperature-monitors and recorders remains to be determined.

Adverse events associated with RFID application for medicines have not yet been reported or identified. Studies showed that the nonthermal effects of RFID radiation remain undetectable in biopharmaceutical products, even after unusually long exposure durations *(61)*. An ideal RFID read environment would be free of metals and liquids, would enforce an air gap between each object's tag, would allow the RFID tags to be flat in a standardized orientation to the antenna, and would be in free air.

6.5.2 Internet of Things

The Internet of Things (IoT) is the process of adding sensors to physical items and connecting the sensors through communications to exchange data. The data is used to create additional value to the users of the physical item along with value to manufacturers, logistics providers, and other stakeholders. Temperature sensors that use such communication methods as Bluetooth, radio frequency, and cellular communication networks are an example of IoT in the pharmaceutical industry. These communication methods enable a real-time understanding of temperature and, in the case of cellular communication networks, a location.

6.5.3 Blockchain

A blockchain is a tamper-proof, distributed database that stores blocks of information for cryptographically bound transactions via peer-to-peer networks. It is an emerging technology that is expected to open up a promising future for the pharmaceutical industry in counterfeit drug prevention, drug distribution, tracking and tracing, safety and security, and temperature control among other functions *(62)*.

The architecture of a blockchain enables users to share validated and updated ledgers for every transaction. Nodes receive copies of each transaction and validate them, synchronizing this information throughout the network. As such, the need for a central validating authority is eliminated. The connections between transactions in separate blocks are linked, which gave rise to the term "blockchain" (see **Figure 6.5.3-1**) *(63, 64)*.

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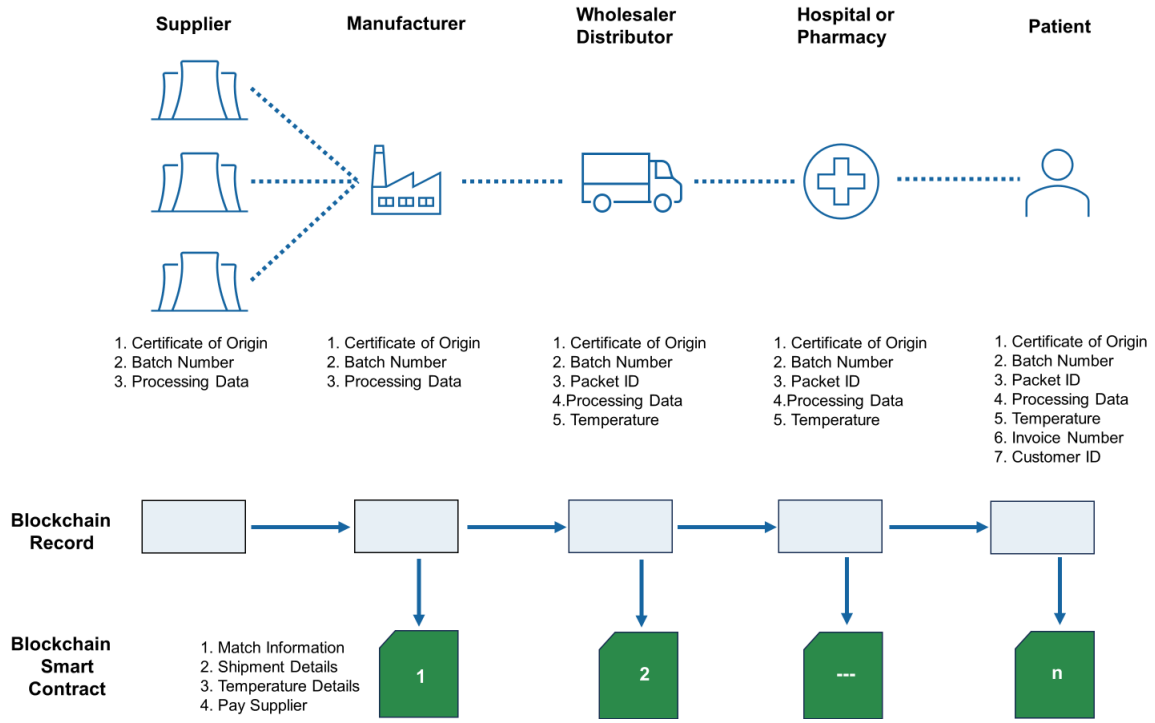

Figure 6.5.3-1 Internet of Things Based Blockchain for Temperature Monitoring and Counterfeit Pharmaceutical Prevention. Rajani Singh, Ashutosh Dhar Dwivedi and Gautam Srivastava. 2020 (64)

6.5.4 Machine Learning

Machine learning as part of artificial intelligence is the combination of data and algorithms that result in a learning process with the goal of producing optimal results. In many cases, the data comes from multiple sources, and with the combined data, additional value is achieved to produce more accurate and optimal answers for the purpose of reducing risk and managing costs. The process of data collection is ongoing and, as more data is added over time, machine learning continues to further optimize solutions. Combining machine learning with IoT produces an immediate understanding of real-time situations that enables users not only to make quicker decisions but, more importantly, to be told what the optimal answers are. The current focus in the pharmaceutical industry is to protect product quality and patient safety. With machine learning and IoT, the industry can optimize existing outcomes and also find more cost-effective solutions.

The last mile is often the most expensive part of the delivery process due to fuel costs and labor expenses. In addition, companies must also factor in the cost of expired and damaged products, failed deliveries, returns, and reverse logistics when planning a shipment. Machine learning algorithms can learn from data and improve over time to optimize last-mile deliveries and reduce costs. Along with the IoT temperature data, additional data can be combined such as weather, traffic delays, airport and seaport delays, and geopolitical events. When data is combined, optimal answers can be achieved to protect the product and also reduce cost. Machine learning can improve last-mile deliveries by predicting customer preferences as well as routing vehicles through traffic patterns and congestion levels, allowing companies to plan their routes accordingly *(65)*. Additional optimization can be achieved by selecting the correct packaging to reduce both packaging and logistics costs.

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6.6 Management of Deviations and Complaints

Any last-mile distribution-related events categorized either as a deviation or complaint should be recorded and appropriately investigated according to written procedure. There should be a risk-based decision process in place to assess product-quality impact and to identify root cause for events with potential product-quality impact *(50)*. All the relevant event data, including any investigation results and CAPAs, should be accessible and ready to share with local regulatory authorities and entities responsible for distribution such as manufacturer, MAH, or logistics service providers.

6.7 Performance Management

From a supply chain perspective, products should be delivered on time, to the right place with the right quality and quantity. Any disturbance should be appropriately managed to circumvent product shortages. Last-mile metrics should promote consistency and accuracy in the distribution process. For each metric, a target and baseline should be determined; the target can be an internal or external benchmark. The baseline allows for analysis against previous performance, while the target allows for analysis against expected performance. Any time a target is not met consistently, or a trend predicts a possible issue, a root cause analysis should be considered, and a corrective action plan should be initiated if appropriate.

Analysis of complaints from shipment recipients provides data on additional shipping and distribution issues, such as carrier service level and accuracy. Organizations can use these metrics to identify vendors that promote accuracy and a high service level within the last mile.

Shipping error rates provide information on the number of packages that reached customers right first time. Analysis of errors may include, but is not limited to, incorrect addresses, incorrect recipients, incorrect quantity, and temperature excursions. Temperature-monitoring with periodic sampling rates during qualification testing is required to ensure appropriate shipping temperature throughout the duration. The data should be analyzed to ensure appropriate temperature ranges can be maintained throughout the shipment and may be input to lane risk assessments.

6.8 Last-Mile Education

Maintaining the quality of temperature-sensitive drugs, vaccines, and biological products requires the consistent application of effective practices by manufacturers, distributors, prescribers, pharmacists, and patients (end users). Pharmacists, nurses, and other healthcare providers are the last interface between the patient and their medication, and they should educate patients regarding the proper storage of products to ensure product integrity at the patient level, as well as the management of unacceptable medications such as recall, rejected and returned products. Implementation and maintenance of such practices requires the significant sharing of information within the supply chain.

Last-mile education is the responsibility of all the involved stakeholders from the manufacturer to the patient and any distribution intermediary. The level of education and training on overall GDP and specific product storage and distribution practices is commensurate with the level of risk associated with the distribution of the product to the patient. Thus, each distribution process should be the subject of a documented risk assessment on all process-critical control points to identify potential time and temperature abuse that could have an adverse effect on product quality. Any risk identified requires mitigation measures, which should be part of the training on a last-mile distribution process.

GDP training should capture all involved stakeholders—product manufacturer, wholesalers, hospital and pharmacy employees, and the patient. Education requirements should be defined as prerequisite to engage in last-mile distribution, while training may be product- or process-specific and depend on a specific role the stakeholder plays in the last-mile supply chain.

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Therefore, stakeholders should develop a specific education and training matrix for the different levels of responsibility from acceptance, transport, and storage to application for the patient or home storage. References used for the training material should capture both international and national regulatory requirements as applicable. Patients need to be informed about access to websites where additional information can be found on last-mile storage and distribution of the product.

Vaccines present a particular distribution challenge within the cold or frozen supply chain. Many vaccines are heat-sensitive and/or freeze-sensitive, and some require specific freezing conditions. Medicinal prescribers annotate their written prescriptions with instructions for the pharmacist to further instruct the patient or caregiver regarding the proper transport and storage of vaccines. Consequently, it is highly recommended that the dispensing personnel become—and remain—thoroughly trained with the transport and storage characteristics of the vaccines they dispense. For example information on the general education, scheduling, transport, and stability of vaccines can be found on the U.S. Centers for Disease Control and Prevention website (http://www.cdc.gov/vaccines). Specific manufacturer-derived information may be obtained from most local medical societies.

The vaccine supply chain and last-mile distribution are still challenging, and closer cooperation of all stakeholders is required. With the recent COVID-19 pandemic vaccine campaigns, several organizations have issued specific guidelines on how to handle, transport, store, and apply such vaccines. Examples include but are not limited to:

1. World Health Organization (WHO) guidance on the shipping of vaccines and specific initiatives related to the COVID-19 pandemic covering various topics such as:
    - **National Deployment and Vaccination Plan**—Covers also country readiness and delivery workstream to disseminate guidance, trainings, tools, and advocacy advice
    - **Vaccine Introduction Readiness Assessment Tool**—Completing the tool can help countries assess their readiness to introduce COVID-19 vaccines, identify process and quality gaps, provide a roadmap, and self-monitor
    - **COVAX Access to COVID-19 Tools (ACT) Accelerator** —A global collaboration project to help accelerate development accelerate the development, production, and equitable access to COVID-19 tests, treatments, and vaccines and enable access to the diverse portfolio of vaccines under development
2. U.S. Centers for Disease Control and Prevention's (CDC) COVID-19 Vaccination Program Interim Playbook for Jurisdiction Operations provides guidance to U.S. states and territories on how to plan and operationalize a vaccination response to COVID-19 within their jurisdictions.

Other international and national organizations have developed similar programs addressing concerns related to the last-mile distribution and application of COVID-19 vaccines to patients.

6.9 Good Distribution Practice Information for Healthcare Professionals (Dispensers)

Healthcare professionals or pharmacies dispensing temperature-controlled drug products must provide effective packaging and instructions for the handling and storage of those products during transit to the final destination. Verbal and/or written instructions, like those below, should be provided in an appropriate language to the patient, relative, or caregiver.

- Educate and train staff on storage requirements for all drug products
- Display required storage temperatures in convenient locations

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- Continuously monitor refrigerator and freezer temperatures, and train responsible persons on the escalation process in case of deviations
- Establish a schedule and procedures to defrost and clean refrigerators; describe the process in a standard operating procedure and train responsible employees accordingly
- Maintain an adequate amount of refrigeration/freezer capacity and manage the capacity against a documented risk assessment
- Establish a schedule to rotate temperature-controlled products, including monitoring expiration dates and train all involved employees on the process
- Verify that drug products within emergency vehicles are properly stored and rotated, ensure that dispatchers and drivers are trained to these procedures, and make sure training verification is in place
- Identify temperature-sensitive products at the time of dispensing and apply appropriate labels to the prescription container according to defined procedure; ensure all involved employees receive adequate training
- Identify for nurses those drugs that are to be stored in hospital floor refrigerators and emphasize the need to promptly return (where applicable for multi-use products) a patient's vial to the refrigerator; ensure proper training is in place and a record of training is kept
- Ensure that pharmacy delivery vehicles and shipment containers meet established criteria for temperature extremes and are designed in accordance with GDP requirements
- Ensure the use of temperature-controlled vehicles, equipment, and packaging is defined and documented, and ensure training records are maintained for all employees
- Provide verbal and written instructions in appropriate language(s) to the patient, relative, friend, or other caregiver; provide additional information after dispensing, as needed, to enhance patient compliance
- Educate patients on how to perform temperature checks of temperature-sensitive products that are delivered to their home
- Provide appropriate information to patients to keep frozen immunization vials frozen until they arrive at their physician's office for injection

6.10 Information to the Patient/Recipient

Patients receiving their prescription drugs should also receive instructions on correct handling and storage. Patients should be aware that keeping their prescription drugs in a temperature-uncontrolled environment (e.g., car) could expose the drugs to temperatures that are too hot or cold. If the drugs require refrigerated storage, patients must store these drugs in the home refrigerator, preferably on the front shelves as the back of the refrigerator may expose drugs to freezing conditions. If the drugs need to be stored at room temperature, patients should put them in a safe, dry place where they will avoid direct sun exposure.

Patients receiving their prescription drugs through a mail-order pharmacy need to be informed of the following:

- Open the packaging immediately and inspect the products for any damages
- Place the product in its appropriate storage temperature

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- Explain to patients that, even though the cool bag may be completely melted, the products are safe as most mail-order pharmacies are qualifying their cold chain solutions with an additional 12 hours after delivery
- Follow instructions on how to dispose of the shipping container and its components (trash, recycling, or composting advice)
- Communicate with a pharmacist about any drug or packaging questions

To ensure that products maintain their quality, effectiveness, and safety through to expiry, data-driven management systems play a vital role in planning, inventory management, demand management, and environment control (e.g., temperature). By applying ALCOA+ principles, data quality provides evidence of correct product storage and transport conditions throughout the supply chain to the patient.

Two of the biggest challenges associated with the pharmaceutical supply chain are temperature-monitoring and counterfeit drug prevention. Many drugs and vaccines remain viable only within a specific range of temperatures. A viable smart healthcare ecosystem could be enabled by applying a real-time shipment temperature-monitoring device connected to the IoT, cloud database(s), and/or block chain technologies. Real-time monitoring may pose new challenges to security and data management, which may require additional resources to manage.

7.0 Cost Management and Sustainability

To ensure product availability with the right quality, effectiveness, and safety, cost-effective measures are applicable across the supply chain, from the manufacturer to the points of dispensing and administration. These measures, which include but are not limited to, inventory management, sustainability programs, and temperature control, should help prevent product shortages and have a positive impact on patient health.

7.1 Inventory Management

7.1.1 Rotating versus Stockpiling

An important criterion for the cost of stockpiles is if they are rotating or static.

Emergency stockpiles for crisis situations: Product shelf life varies by the type of product. Stockpiles can be partially dynamic, but also require holding stocks that do not rotate if they are dedicated to a specific emergency or to crisis situations and therefore need to be held as part of a static stockpiling approach (e.g., stockpiling of masks prior to the COVID-19 pandemic or, more recently, iodine tablets). If stocks are not managed on a rotating basis with "normal" inventory, there is a high risk of expiration. This necessitates their disposal, generating unnecessary additional costs, creating a complete loss of the initial investment, and triggering a negative environmental impact. A static stockpile, with no stock rotation, might therefore result in a large volume of expired stock that needs to be replenished, causing additional costs for their creation, maintenance, and the turnover of the stockpile. Who covers the cost of expired products and who pays for their disposal should be defined by contract.

Increasing buffer stocks for critical or essential medicines: This could be integrated via rotation into pharmaceutical full-line wholesaler/full-service healthcare distributor rolling stocks thereby reducing the risk of products expiring and could be carried out on a FEFO basis, i.e., dynamic stockpiling. Naturally, the shelf life of products differs according to the type of product; however, buffer stocks can be organized in a way that the products are continuously used and replenished. In that case, the length of storage, expiry date of products, and potential safe disposal of unwanted medicines need to be considered. Increased buffer stocks held at the wholesale level need to be purchased from the pharmaceutical industry, which increases the capital costs. This is one of the main

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reasons, other than available warehouse space, why the establishment of larger stockpiles of more than two weeks at the level of pharmaceutical full-line wholesalers and full-service healthcare distributors is highly challenging. This can be attributed to insufficient margins, while also holding a prefinancing function for the healthcare supply chain.

7.1.2 Organizational and Administrative Reflections

Additional administrative and logistics costs may apply to storage, based on the conditions and to whom the safety stocks should be delivered. Costs are higher the more frequently the list of products to be stocked changes because it requires a greater effort to administer the stocks. A decision should also be made as to what condition the emergency stocks will be supplied or released to the market, how they should be rotated (e.g., FEFO), and how the rationalization or allocation system would be organized with respect to market deliveries to ensure equitable access.

7.2 Warehouse Storage and Temperature Control

The inventory level, buffer-stock determination, and size of individual packs should be considered when establishing the right size of warehouse. Choosing the right size of the warehouse is key to minimizing the cost of energy that will be required for temperature control. The cost of energy is one of the highest fixed costs of an organization. The use of engineering solutions can be helpful in saving energy. For example, mineral wool batting, polyurethane (PUR), polyisocyanurate (PIR) and polystyrene (EPS), used as insulation to reduce energy consumption. Also, the application of solar panels may be used to generate power and reduce energy costs.

7.3 Waste Stream for Packaging Components and Drug Product

Sections 7.3–7.5 provide a non-exhaustive overview of various global directives and waste stream management for packaging components used to distribute temperature-sensitive products in the last mile. Packaging components, including ice bricks, boxes, and isolated containers, may generate waste for each container shipped (e.g., 20 Kg per container). Depending on the time period, thousands of waste materials can be generated, hence the importance of studying the reuse or recycling for this business.

7.3.1 Recycling and Regulations

Most national and supranational entities have legislative guidelines in effect governing the proper post-transit disposal of pharmaceutical products and devices. Some also cover the packaging materials used during transport. Currently, the "extended producer responsibility" or Extended Product Responsibility (EPR) program, has been instituted in several countries. The concept behind EPR implies that everyone along the product chain shares responsibility for the environmental impact of packaging. That includes the upstream considerations of material selection and manufacturing processes to the downstream effects of use, waste management, and disposal. However, this approach largely sidesteps the issue of who is directly responsible for the waste and disposal of post-consumer packaging. Some corporations have instituted their own EPR programs, but those are generally confined to the return of specific product components (e.g., printer or copier ink cartridges).

European Union—Directive 94/62/EC on packaging and packaging waste forms the basis of the EU's position toward packaging issues; this directive was amended by Directive 2004/12/EC to include states newly admitted to the EU. Further information is available from the European Committee for Standardization (http://www.cen.eu/cenorm/homepage.htm) and the European Organization for Packaging and the Environment (http://www.europen.be). The EU GDP does not directly address the impact of packaging on the environment, however, the EU has adopted "extended producer responsibility" as its public policy. This requires packaging producers to be fiscally and physically responsible for their products at the post-consumer stage of their lifecycle. This approach

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creates strong incentives for producers and manufacturers to incorporate external-packaging costs, like waste managements, into their designs from their inception. Packaging waste can be "taken back" or managed through reuse, recycling, or remanufacturing. Optionally, the responsibility may be delegated to a third party to act as a producer-responsibility organization. Producers pay certain fees and taxes to governmental and nonprofit entities to cover the reclamation and disposal costs. For example, the Green Dot Program operates in 20 EU member countries where a green dot on a product's packaging indicates it complies with packaging regulations. Manufacturers purchase licenses to participate in the Green Dot program, which collects and recycles or otherwise disposes of the packaging on behalf of the license-holders. The EU's EPR program has recently expanded to automobiles and electronics in many member states.

United States—In the United States, the sole federal EPR program involves the take-back of rechargeable (nickel-cadmium) batteries, in which the producers of the batteries instituted a national take-back program to avoid variable individual state legislation. Most shipments of pharmaceuticals to the consumer, particularly biologicals, use EPS/foam packaging, which usually winds up in landfills. The Alliance of Foam Packaging Recyclers has posted a resource document on their website (http://www.epspackaging.org) listing available recycling/reclamation sites in the United States for EPS packaging. At the present, a "true" national EPR public policy and program remains an unmet need.

Japan–Japan's packaging directives are part of the environmental Japanese Industrial Standards, which take their cue from *JIS Q 0064: Guide for the Inclusion of Environmental Aspects in Product Standards* ***(66)***. More information about the Japanese Industrial Standards Committee can be found at http://www.jisc.go.jp/eng/.

Brazil–Resolution SMA 45/2015 of 23 June 2015 establishes the Terms of Commitment for the Reverse Logistics of Household Medicines for Human Use, Expired or Disused, and their Packaging, according to the standardized model made available by the State Secretariat for the Environment (SMA) and the Environmental Company of the State of São Paulo (CETESB). (1) Consumers must dispose the expired or disused medicines and their packaging at one of the receiving points implanted in pharmacies and drugstores; (2) Drugstores and pharmacies established as receiving points are obliged, at their expense, to make available and maintain their container dispensing establishments, in order to promote the implementation of the reverse logistics system of medicines discarded by consumers and their packaging; (3) Distributors of medicines are obliged to collect containers containing the medicinal products discarded by the consumer from the primary storage point (pharmacies and drugstores) to the secondary storage point, provided that this collection can be carried out by the same vehicle and at the same time as the delivery of medicinal products to the primary point; and (4) Manufacturers and importers of medicinal products shall be obliged to carry out, at their own expense, by their own means or by means of third-party contracts, the transport of medicinal products discarded by consumers from secondary storage points to the final treatment sites and environmentally appropriate final disposal ***(67, 68)***.

7.3.2 Reverse Logistics

Reverse logistics encompasses all operations related to the upstream movement of products and materials. It is the process of moving goods from their typical destination for the purpose of capturing value or for proper disposal. Remanufacturing and refurbishing activities also may be included in the definition of reverse logistics. Two types of reverse logistics can be considered in the pharmaceutical environment:

Closed-Loop Transit

Many wholesalers in Europe and the United States use this system. Most of these companies use their own trucks to deliver to the end users, where they can then pick up old packaging and put it back into the distribution channel or recycle the components.

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Return of Packaging

Some manufacturers and distributors employ this method, wherein a prepaid return freight sticker is attached or included with the original shipment to the end user. Once the end user removes the medicinal product, the package can be sent back to the distributor or manufacturer. This method is also used to return expired or mishandled medicinal products.

7.4 Establishing Lowest-Cost Temperature-Control Solution for Shipment

Freight is an important cost factor to be considered when establishing a solution for temperature-controlled shipment. Some examples of other factors that play a role in the decision are the mode of transportation, the type and weight of effective thermal packaging, the amount of time spent in transport, and the environmental impact of the packaging.

There are different kinds of isolated passive containers available, which may be made of EPS, extruded polystyrene (XPS), polyurethane (PUR), or vacuum insulated panels (VIP), and refrigerants made of water and/or phase-change material (PCM). Passive thermal solutions composed of VIPs and PCM are typically more expensive than more traditional solutions of EPS boxes and water-based refrigerants. Traditional solutions are usually physically larger and heavier for the same quantity of product to be shipped, which results in higher freight costs. The freight costs are generally greater than the temperature-control solution costs, but this solution can significantly impact the total cost of shipment.

Air freight is generally more expensive than ocean freight or road freight. Transport by road in refrigerated trucks is typically less expensive than shipping passive-thermal solutions by air, but the time it takes to reach the destination and the ability to plug the truck into an electrical power supply when the active cooling unit on the truck is turned off (e.g., on a ferry) are necessary considerations.

Even though air freight is the most expensive mode, sometimes it is the best solution or may be the only solution when considering the time available for shipment or accessibility of the destination, for instance, island locations or other difficult-to-reach areas without passable roads. Passive-thermal containers are generally less expensive than the active containers shipped by air, but their performance time is limited, and they may not provide sufficient protection before reaching the destination.

Both passive and active transport equipment is designed for a defined amount of time under specific anticipated external temperature conditions as used in the qualification studies. It is important to consider these factors when evaluating the suitability of such packaging for its intended use, as well as knowing the routing specifics, risks associated with critical control points in the process, and the overall transit time from origin to destination.

Reuse is another way to reduce the cost of passive-thermal solutions. The reuse process should be qualified and have a documented quality control process for the returned solutions, including cleaning and inspections to ensure the used system will provide acceptable performance for its intended use.

7.5 Cost of Failure

Establishing a robust shipment process with a qualified solution is an investment made to avoid losses due to quality issues. The cost associated with damage, recalls, returned goods, and rejection due to temperature-control failure is high. The cost can be even higher if the damage is to patient safety and health because of a failure in temperature-monitoring and appropriate quality controls. Any of these circumstances can damage the company brand and image.

Mistakes in purchase orders, invoices, selecting materials picking and separation, address and customers should be avoided. The result of mistakes impacts products and may lead to rejection due

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to limited or absence of supporting stability data. Insuring high-value shipments because of the high risks associated is common.

7.5.1 Business Contingency Plan

The goal of risk-based planning is to focus on preparing the organization for a multitude of risks, including man-made and natural disasters. Some points to consider when developing an emergency plan are:

- Prioritize the operations and responsibilities to guide a disaster response in advance
- Consider alternative locations for operations such as call centers, offices, warehouses, product processing, transportation operations, or equipment
- Back up data and IT infrastructure in a remote location outside the anticipated disaster impact zone
- Consider ways to reroute transport operations and protect or remove a product that is in an impact area
- Prioritize key activities and responsibilities and designate personnel who will be responsible for these activities if regular personnel are unavailable
- Plan for the cascading effects of shifting responsibilities if multiple persons are unavailable, and reorganize remaining personnel in response to the disaster
- Establish local-level, site-specific business continuity plans tailored to individual business centers and individual customers and products, as needed
- Prevent man-made disasters to the greatest extent practical based on the situation (the Customs-Trade Partnership Against Terrorism guidelines produced by the U.S. Customs and Border Protection provides an example of resources to guide this effort)
- Implement reporting systems and tools, alarms, and other automated monitoring systems to speed notification of an incident and enable a timely response
- Establish a response plan and designate a responsible person to coordinate appropriate intervention activities
- Apply the principles of quality assurance to monitor, measure, and continuously improve disaster-intervention results

8.0 Summary

The intent of this technical report is to convey the importance of proper information, training, and communication between handoffs as vital to maintaining medicinal product integrity in the pharmaceutical supply chain, especially in the last mile. Knowledge of the challenges encountered along the last mile allows for improvements in future guidance documents, products, and services to the temperature-controlled supply chain community.

The last mile of delivering temperature-controlled pharmaceutical products has many challenges to manage including, but not limited to, the complexity of pharmaceutical supply chain networks, various pharmaceutical product characteristics and temperature sensitivities, weather conditions, sourcing of services and temperature-controlled solutions, use of new technologies and communication platforms, developments in new regulations, execution of cost-effective and sustainable measures, and education in product handling. Sharing information about conditions and practices for the storage and transport of product across all supply chain partners is critical to maintaining product quality, applying

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appropriate measures, and preventing temperature excursions in the supply chain, including the last mile. When all supply-chain partners involved collaborate closely, the last-mile partners can deliver products on time, to the right place, with the right quantity and quality to the end user.

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